

# Midterm Review

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## Abstract

We study a risk-neutral principal who may review Gaussian cumulative output at an endogenous date and condition bounded, limited-liability compensation and continuation on performance. The optimal review menu has two, three, or four ordered regions: a bad stop, low-rent continuation, high-rent continuation, and possibly a good stop with the maximal prize. Review timing balances the informativeness of output against continuation rents available as incentive capital. Arbitrarily early reviews reduce value, whereas sufficiently tight feasible payment caps make an interior review strictly optimal. The model identifies purely incentive-driven termination and rent recycling in a canonical net-output agency setting.

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# 1 Introduction

Interim performance reviews are commonly used to decide whether an agent should be retained, how demanding future targets should be, and whether exceptional early performance should be rewarded immediately. These decisions are usually analyzed separately. This note studies them jointly in a canonical moral-hazard environment. A risk-neutral principal receives terminal output net of compensation, a risk-neutral agent chooses phase-level binary effort, cumulative output follows a Brownian motion, and compensation is subject to limited liability and a hard upper bound. The principal may commit to a single public review at an endogenous date  $\tau$  and condition the post-review opportunity on the realized state.

Without a review, bounded payments lead to the familiar likelihood-ratio bonus: the agent receives  $M$  only when terminal output exceeds a threshold. The dynamic problem becomes substantially richer because continuation itself creates rents. Those rents need not be treated simply as an agency loss. By assigning more valuable continuation opportunities to better review outcomes, the principal recycles unavoidable future rents into incentives for first-phase effort. The review therefore allocates what may be called *incentive capital*.

The first main result is an ordered performance ladder. For a fixed review date, the post-review full-effort frontier is a line segment between a minimum-rent contract and a maximum-rent contract. Strict MLRP at the review and linearity of the principal's assignment problem imply that only the endpoints of this frontier and the endpoints of the no-effort frontier are used. Depending on parameters, the optimal review scheme has two, three, or four nonempty regions. Poor outcomes receive a *bad stop*, termination, with zero payment. Adequate outcomes receive low-rent continuation under a demanding terminal bonus. Better outcomes receive high-rent continuation under an easier bonus. When first-phase incentives are especially costly, exceptional outcomes receive a good stop: the principal pays  $M$  immediately at the time of the review and sacrifices second-phase production. Because the productivity of future effort is independent of the review state, both stopping regions are purely incentive driven rather than responses to bad or good project types.

The second main result concerns review timing. Waiting longer separates the review-state distributions under work and shirking and therefore improves the statistical quality of performance measurement. But delay also shortens the continuation phase, reducing both its productive surplus and the rents that can be awarded as prizes for first-phase effort. The optimal review date  $\tau$  balances information against incentive capital. This is a genuinely dynamic tradeoff: the principal chooses not only how to reward performance, but also how much future agency rent to create for use as a current incentive instrument.

In the Brownian baseline, the endpoint behavior makes this tradeoff especially sharp. The right derivative of the review value at  $\tau = 0$  is minus infinity. An arbitrarily short probationary phase is therefore harmful for incentive provision, and every strictly optimal review is bounded away from the initial date. Short probationary periods observed in practice must therefore be justified by motives outside pure moral hazard or by a monitoring technology with hard short-horizon evidence, as in Section ???. At the opposite endpoint, the left derivative at  $\tau = T$  is explicit. When the

payment cap is sufficiently tight, carving out even a short continuation phase has first-order value because it creates scarce continuation rents. If the no-review full-effort benchmark is nonnegative at the feasibility boundary, every sufficiently tight feasible cap makes an interior review strictly optimal. With a loose cap, by contrast, the terminal contract can provide incentives directly and the marginal value of a continuation phase disappears.

The analysis proceeds as follows. Section 3 derives the no-review benchmark, and Section 3.2 constructs the post-review continuation frontier. Sections 4 and 4.3 formulate pre-review incentives and characterize the ordered menu and the boundaries between its two-, three-, and four-region forms. Section 5 derives the review value, endpoint derivatives, and the tight-cap result. Section 6 relates these mechanisms to work on bounded payments, interim evaluation, termination, and feedback design.

## 2 Environment

A risk-neutral principal (she) contracts with a risk-neutral agent (he) over a finite horizon  $[0, T]$ . Output  $X_t$ , the state, evolves as

$$dX_t = e_t dt + dW_t, \quad X_0 = 0,$$

where  $W$  is a standard Brownian motion and  $e_t \in \{0, 1\}$  is effort. We write  $\Phi(\cdot)$  for the standard normal distribution and  $\phi(\cdot)$  for its density.

The contract fixes at date 0 a single review date  $\tau \in (0, T]$ .

**Assumption 1** (Observability of the State). *(i) The state is publicly observed at  $\tau$  and at  $T$ .*

*(ii) No-one, including the agent, observes  $x_t$  at any other time.*

The review splits the horizon into two phases,

$$P_1 = (0, \tau], \quad P_2 = (\tau, T], \quad \text{of lengths } \tau \text{ and } T - \tau.$$

The case  $\tau = T$  is the *no-review benchmark*: the second phase is empty and  $X_T$  is the only signal.

**Assumption 2** (Phase-level private effort). *Effort is constant within each phase. The agent privately chooses  $a_1 \in \{0, 1\}$  at date 0 and, having observed  $X_\tau$ , privately chooses  $a_2 \in \{0, 1\}$  at date  $\tau$ ; effort is  $e_t = a_1$  on  $P_1$  and  $e_t = a_2$  on  $P_2$ . A strategy is a pair  $(a_1, S)$  with  $S : \mathbb{R} \rightarrow \{0, 1\}$  giving  $a_2 = S(X_\tau)$ .<sup>1</sup>*

The agent commits at the start of a phase to work throughout it or shirk throughout it, and cannot revise until the next observation. Consequently the two phase increments are independent

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<sup>1</sup>\*\*\***YG: TBA**\*\*\* Assumption 2 is relaxed in Section ??, where the agent chooses  $e_t \in \{0, 1\}$  at each instant, so that a phase of length  $h$  delivers any total effort  $\eta = \int e_t dt \in [0, h]$  rather than only  $\{0, h\}$ .

given  $(a_1, a_2)$ , with

$$X_\tau \sim N(a_1\tau, \tau), \quad X_T - X_\tau \sim N(a_2(T - \tau), T - \tau).$$

Working through a phase of length  $h$  costs the agent  $ch$ , so total effort cost is  $c[a_1\tau + a_2(T - \tau)]$  with  $c \in (0, 1)$ . The same flow cost applies in both phases, which matters because  $\tau$  will later be chosen endogenously. After observing  $X_\tau$  and  $x_T$  the principal makes a measurable payment  $w(X_\tau, X_T)$  to the agent.

**Assumption 3** (Limited liability and finite compensation). *The payment function satisfies*

$$w : \mathbb{R}^2 \rightarrow [0, M], \quad M \in (0, \infty).$$

The lower bound of 0 is the agent's limited liability, and the upper bound  $M$  is a finite cap; i.e., the maximum payment the principal can make.<sup>2</sup>

The principal receives ex post payoff  $X_T - w$ , and the agent receives  $w - c[a_1\tau + a_2(T - \tau)]$ . The principal chooses the review date  $\tau$  and the payment function  $w$  to maximize her expected payoff. Under the no-review benchmark  $\tau = T$  a contract is simply  $w(X_T) \in [0, M]$ .

### 3 Contracting over a single phase

Fix a phase of length  $h > 0$  at the end of which output is observed and payment is made, with no further review. Two cases of the model have this form,  $h = T$  and  $h = T - \tau$ ; in the latter, translation invariance of the technology lets a continuation contract be written as a function of the terminal increment  $Y = X_T - X_\tau$ . Solving the phase problem once therefore delivers both together.

Under work  $Y \sim N(h, h)$  and under shirking  $Y \sim N(0, h)$ , with densities

$$f_j^h(y) = \frac{1}{\sqrt{h}} \phi\left(\frac{y - jh}{\sqrt{h}}\right), \quad j \in \{0, 1\},$$

and working costs the agent  $ch$ . A contract is a measurable  $w : \mathbb{R} \rightarrow [0, M]$ ; it implements work if

$$\mathbb{E}_1[w] - \mathbb{E}_0[w] = \int w[f_1^h - f_0^h] \geq ch, \quad (\text{IC}_h)$$

where  $\mathbb{E}_j[w] = \int w f_j^h$ . Let  $\mathcal{F}_h$  denote the set of contracts satisfying  $(\text{IC}_h)$ .

The goal of this section is to describe the set of rent–payoff pairs a single phase can deliver. That set does double duty. At  $h = T$  it is the *no-review benchmark*: its best point for the principal is the most she can earn when output at  $T$  is her only instrument. At  $h = T - \tau$  it is the continuation problem after a review at  $\tau$ , and hence the menu of continuation utilities she can promise there; Section 4 shows that assigning different points of this set to different review outcomes is exactly

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<sup>2</sup>All results generalize to a setting with finite lower and upper bounds on payments  $-\infty < \underline{w} < \bar{w} < +\infty$ , where  $M \equiv \bar{w} - \underline{w}$ . We set  $\underline{w} = 0$  as is standard in settings of limited liability.

how she buys first-phase effort. One object therefore answers two questions: what a phase is worth on its own, and how much of that value can be converted into incentives for the phase before it.

### 3.1 Threshold contracts

We begin by showing that it is without loss to focus on *threshold contracts*: pay the maximum  $M$  when output clears some bar  $k$ , and nothing below it. Any contract can be matched by a threshold contract costing the principal the same when the agent works, because  $\mathbb{P}\{Y \geq k \mid e = 1\}$  falls continuously and strictly from one to zero as  $k$  rises. The following lemma says that this substitution never destroys incentives; it strengthens them.

**Lemma 1** (Threshold reduction). *Let  $w : \mathbb{R} \rightarrow [0, M]$  be any contract and let  $\tilde{w} = M\mathbf{1}\{y \geq k\}$  be the threshold contract with the same expected cost under work,  $\mathbb{E}_1[\tilde{w}] = \mathbb{E}_1[w]$ . Then*

$$\mathbb{E}_0[\tilde{w}] \leq \mathbb{E}_0[w],$$

*with equality only if  $w = \tilde{w}$  almost everywhere. Hence  $\mathbb{E}_1[\tilde{w}] - \mathbb{E}_0[\tilde{w}] \geq \mathbb{E}_1[w] - \mathbb{E}_0[w]$ , so  $w \in \mathcal{F}_h$  implies  $\tilde{w} \in \mathcal{F}_h$ .*

Lemma 1 is a statement about the price of incentives. Raising the payment slightly near the outcome  $y$  raises expected compensation at rate  $f_1^h(y)$  and raises the incentive spread at rate  $f_1^h(y) - f_0^h(y)$ , so the cost of one unit of incentive delivered at  $y$  is

$$\frac{f_1^h(y)}{f_1^h(y) - f_0^h(y)} = \frac{1}{1 - \ell^h(y)}, \quad \ell^h(y) \equiv \frac{f_0^h(y)}{f_1^h(y)} = \exp\left(\frac{h}{2} - y\right). \quad (1)$$

The ratio  $\ell^h$  records how much more likely the outcome is under shirking than under work, and it is strictly decreasing: high output is evidence of effort, and the higher the output the stronger the evidence. Incentives are therefore cheapest at the top of the distribution, and a principal who can pay at most  $M$  at any one outcome buys them there first. Below  $h/2$  the ratio exceeds one, so paying *lowers* the spread; the bar will always be set above  $h/2$ .

Let  $k$  be the bar and index it by the *normalized bar*

$$z = \frac{h - k}{\sqrt{h}},$$

the number of standard deviations by which the bar falls short of expected output under work. A larger  $z$  is a lower and hence easier bar, cleared with probability  $\Phi(z)$  by an agent who works. For the threshold contract  $w = M\mathbf{1}\{y \geq k\}$ ,

$$\mathbb{E}_1[w] = M\Phi(z), \quad \mathbb{E}_0[w] = M\Phi(z - \sqrt{h}), \quad \mathbb{E}_1[w] - \mathbb{E}_0[w] = M\Delta(h, z), \quad (2)$$

where

$$\Delta(h, z) = \Phi(z) - \Phi(z - \sqrt{h}), \quad \bar{\Delta}(h) = \max_z \Delta(h, z) = 2\Phi\left(\frac{\sqrt{h}}{2}\right) - 1. \quad (3)$$

The index  $\Delta(h, z)$  is the increase in the probability of clearing the bar when the agent works rather than shirks, so it measures the incentive power of a threshold-contingent reward;  $\bar{\Delta}(h)$  is the most such a reward can deliver, attained at the bar  $k = h/2$  lying halfway between expected output under shirking and under work. By (2) a threshold contract implements work exactly when  $\Delta(h, z) \geq ch/M$ .

**Lemma 2** (Feasibility). *(i)  $\mathcal{F}_h \neq \emptyset$  if and only if  $\bar{\Delta}(h) \geq ch/M$ .*

*(ii) The ratio  $\bar{\Delta}(h)/h$  is strictly decreasing on  $(0, \infty)$ . Consequently, if  $\bar{\Delta}(T)/T > c/M$  then  $\bar{\Delta}(h)/h > c/M$  for every  $h \in (0, T]$ .*

*(iii) Suppose  $\bar{\Delta}(h) > ch/M$  and let  $z_D(h) < \sqrt{h}/2 < z_E(h)$  be the two roots of  $\Delta(h, z) = ch/M$ . Then the threshold contracts implementing work are exactly those with  $z \in [z_D(h), z_E(h)]$ .*

Part (i) is the statistical limit of bounded incentives: work is implementable if and only if the greatest leverage a capped reward can generate covers the cost of effort per unit of cap. Part (ii) says the binding case is the longest phase, so feasibility need only be checked at the full horizon. Figure 1b plots the ratio: it falls from  $+\infty$  to zero, so it crosses  $c/M$  once, at a horizon  $h^*$ , and full effort is implementable on a phase of length  $h$  exactly when  $h \leq h^*$ . We maintain throughout that the model's horizon lies below that threshold.

**Assumption 4** (Effort feasibility). *The parameters satisfy*

$$\frac{\bar{\Delta}(T)}{T} = \frac{1}{T} \left[ 2\Phi\left(\frac{\sqrt{T}}{2}\right) - 1 \right] > \frac{c}{M}. \quad (\text{AS})$$

Part (iii) is what the rest of the section rests on. Because  $\Delta(h, \cdot)$  is unimodal, the requirement  $\Delta(h, z) \geq ch/M$  carves out a closed interval of admissible bars: the incentive constraint binds at either endpoint and is slack strictly between them. We write

$$K_D(h) = h - z_D(h)\sqrt{h} > K_E(h) = h - z_E(h)\sqrt{h}$$

for the two extreme bars and

$$w_D^h(y) = M \mathbf{1}\{y \geq K_D(h)\}, \quad w_E^h(y) = M \mathbf{1}\{y \geq K_E(h)\}$$

for the corresponding contracts:  $w_D^h$  sets a *difficult* standard,  $w_E^h$  an *easy* one. They generate exactly the same incentive to work, since the incentive constraint binds at both. They differ in price. Figure 1a displays the construction.

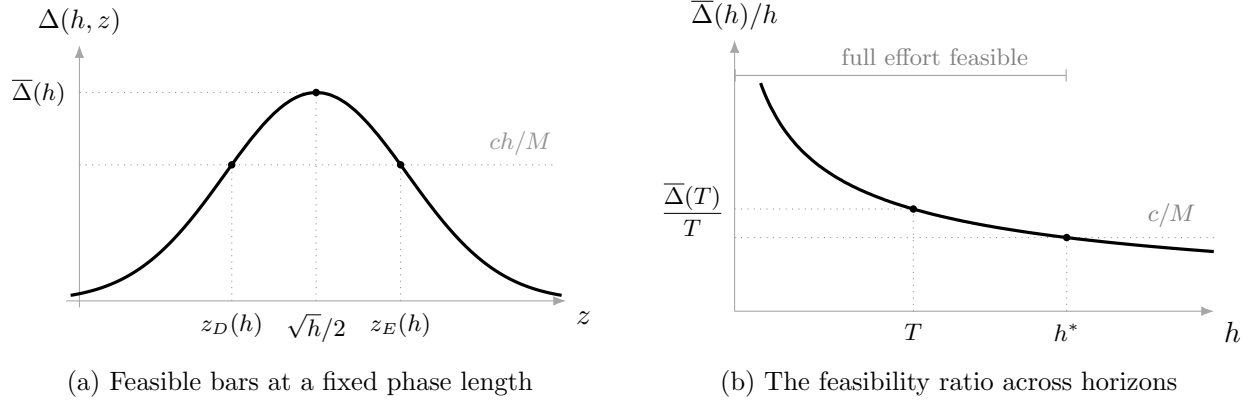


Figure 1: Feasibility of full effort

### 3.2 The optimal contract and the rent–payoff frontier

For  $w \in \mathcal{F}_h$  define the agent’s rent and the principal’s expected payoff,

$$r(w; h) = \mathbb{E}_1[w] - ch, \quad u(w; h) = h - \mathbb{E}_1[w], \quad (4)$$

so that total surplus is  $s(w; h) = r(w; h) + u(w; h) = (1 - c)h$ , and every full-effort contract lies on the line  $r + u = (1 - c)h$ . Effort is undistorted whenever work is implemented, total surplus is the first best, and rent and payoff are pure transfers along that line. The only question is which rents are attainable.

Because  $r$  is an increasing affine function of  $\mathbb{E}_1[w]$ , this is the question of how cheap and how expensive an incentive-compatible contract can be. Both answers follow from Section 3.1. Lemma 1 holds the cost fixed and improves incentives, so it speaks to every expenditure level and not merely to the cheapest one: if some contract costing  $C$  implements work, the threshold costing  $C$  implements it too. The attainable costs are therefore unchanged if we restrict attention to threshold contracts, and by Lemma 2(iii) the admissible thresholds are exactly  $z \in [z_D(h), z_E(h)]$ , over which  $\mathbb{E}_1[w] = M\Phi(z)$  is strictly increasing.

**Proposition 1** (The full-effort frontier). *Fix  $h > 0$  with  $\bar{\Delta}(h) > ch/M$ .*

(i) *The cheapest and the most expensive contracts in  $\mathcal{F}_h$  are  $w_D^h$  and  $w_E^h$  respectively, each unique up to a set of Lebesgue measure zero, and  $(IC_h)$  binds at both.*

(ii) *Their rents are*

$$r_j(h) = M\Phi(z_j(h)) - ch = M\Phi(z_j(h) - \sqrt{h}), \quad j \in \{D, E\}, \quad (5)$$

*and satisfy  $0 < r_D(h) < r_E(h) < M - ch$ .*

(iii) The attainable rents are exactly  $\mathcal{R}_h = [r_D(h), r_E(h)]$ , so the principal's full-effort frontier is

$$U_h(r) = (1 - c)h - r, \quad r \in [r_D(h), r_E(h)]. \quad (6)$$

The frontier is a segment of slope  $-1$ , and the principal's preferred point on it depends on what she is doing with the phase. If she values only her current payoff she picks its left endpoint.

**Corollary 1** (The payoff-maximizing contract). *The contract maximizing the principal's expected payoff over  $\mathcal{F}_h$  is the difficult standard  $w_D^h$ , which delivers*

$$u_D(h) = h - M\Phi(z_D(h)), \quad r_D(h) = M\Phi(z_D(h) - \sqrt{h}). \quad (7)$$

Both extreme contracts buy the same effort; they differ only in what they concede. The difficult standard pays  $M$  rarely under work and even more rarely under shirking, and because the incentive constraint binds, the rent the agent collects is exactly what a shirker expects to be paid. The easy standard pays more often under both actions, with the same difference  $ch/M$  between the two payment probabilities, and so leaves the larger rent  $r_E(h)$ . Limited liability is what makes the rent positive at all: the principal cannot fine the agent for the low realizations that occur even under shirking, so she cannot recover what a lucky shirker collects.

### 3.3 The no-review benchmark

Setting  $h = T$  recovers the benchmark, and no separate argument is needed: by Corollary 1 the optimal no-review contract is  $w^{\text{NR}} = M\mathbf{1}\{X_T \geq K_D(T)\}$ , and the benchmark rent and value are simply the left endpoint of the frontier,

$$R^{\text{NR}}(h; c, M) = r_D(h), \quad V^{\text{NR}}(h; c, M) = u_D(h) = (1 - c)h - R^{\text{NR}}(h; c, M). \quad (8)$$

The entire agency cost is the rent. The principal is never obliged to implement work: offering  $w \equiv 0$  induces shirking and yields zero, so her value over a phase of length  $h$  is  $\max\{0, V^{\text{NR}}(h; c, M)\}$ . We assume the second term wins at the horizon of the model.

**Assumption 5** (Profitable full effort). *The parameters satisfy*

$$V^{\text{NR}}(T; c, M) = T - M\Phi(z_D(T)) > 0. \quad (\text{AS}')$$

Assumption 5 is imposed at  $h = T$  and, unlike feasibility, does not descend to shorter phases. Lemma 2(ii) guarantees that full effort remains implementable on every subinterval, but implementable is not profitable: as  $h \downarrow 0$  the difficult root  $z_D(h)$  drifts to  $-\infty$  only logarithmically, so expected compensation  $M\Phi(z_D(h))$  falls at rate  $\sqrt{h}$  up to a logarithmic factor while expected output falls at rate  $h$ . Hence  $V^{\text{NR}}(h; c, M) < 0$  for all sufficiently small  $h$ : over a short enough phase the rent required to buy effort exceeds everything the effort produces. This is the same force that makes arbitrarily early reviews harmful in Section 5.



Finally, the payment cap is what makes rents unavoidable, and therefore what makes a review worth having.

**Lemma 3** (The payment cap and rents). *Fix  $h \in (0, T]$ . On the feasible range  $M > ch/\bar{\Delta}(h)$ , the rent  $R^{\text{NR}}(h; c, M)$  is strictly decreasing in  $M$ , and*

$$\lim_{M \rightarrow \infty} R^{\text{NR}}(h; c, M) = 0, \quad \text{so} \quad M\Phi(z_D(h)) \downarrow ch.$$

Without a cap the principal could place an arbitrarily large reward on an arbitrarily extreme upper-tail realization, leaving the agent almost no rent. A finite cap prevents that concentration and forces her to concede a positive rent. An interim review is valuable precisely because these otherwise unavoidable continuation rents can be made contingent on earlier performance and recycled as incentives for earlier effort. Since the cap always binds under optimal compensation, the model is equivalent to one in which the principal holds a single indivisible prize worth  $M$  — a promotion, say, or an employment guarantee.

### 3.4 The review action set and its extreme points

Now take  $h = T - \tau$  to be a continuation phase and ask what the principal can offer at the review. Describe an action by the pair  $(r, s)$  of continuation utility and *total* continuation surplus.

Proposition 1 gives the actions that induce continuation effort. Effort is undistorted whenever it is implemented, so total surplus is the first best  $(1 - c)h$  whichever admissible standard is used, and the attainable rents are exactly  $[r_D(h), r_E(h)]$ . The remaining possibility is to induce none: if the agent shirks in the continuation and is promised  $r$ , the principal's incremental payoff is  $-r$ , implemented by the constant contract  $w \equiv r$ , so total surplus is zero for every  $r \in [0, M]$ . Terminating at the review and continuing with zero effort are equivalent, since the expected future increment is zero either way. The two action sets are therefore horizontal segments,

$$\mathcal{A}_0(h) = [0, M] \times \{0\}, \quad \mathcal{A}_1(h) = [r_D(h), r_E(h)] \times \{(1 - c)h\}, \quad (9)$$

with  $\mathcal{A}(h) = \mathcal{A}_0(h) \cup \mathcal{A}_1(h)$ , drawn in Figure 2.<sup>3</sup>

The picture separates the two things a continuation action does. Its height is the surplus that continuation effort creates,  $(1 - c)h$ , and does not depend on which standard is used. Its width is the rent the principal must concede: the full interval  $[0, M]$  if she gives up on second-phase effort, but only the subinterval  $[r_D(h), r_E(h)]$  if she insists on it. Inducing effort therefore raises total surplus at the cost of losing the extremes of the rent range, which is why the stopping actions survive as instruments even though they destroy output.

The four vertices of the trapezoid are the only actions that will ever be used. Two involve no

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<sup>3</sup>The action set is the pair of segments in (9), not the trapezoid they span. The two have the same support function, so nothing below turns on the distinction; the trapezoid is the relevant set if public randomization is available at the review.

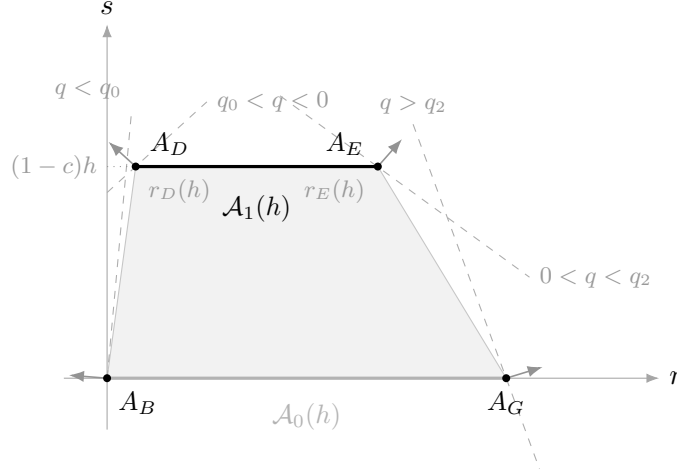


Figure 2: The review action set in rent–surplus space, and the four supporting lines

continuation effort, the *bad stop* and the *good stop*,

$$A_B = (0, 0), \quad A_G = (M, 0),$$

and two involve full continuation effort under the difficult and the easy standard,

$$A_D = (r_D(h), (1 - c)h), \quad A_E = (r_E(h), (1 - c)h).$$

The bad stop terminates without payment; the good stop awards the maximal payment after exceptionally strong first-phase performance but sacrifices second-phase production. The four are ordered by the rent they promise,

$$0 < r_D(h) < r_E(h) < M. \quad (10)$$

Section 4 will value an action by a linear functional  $s + qr$ , in which one unit of surplus is worth one unit and one unit of rent is worth  $q$  on net. Such an objective has gradient  $(q, 1)$  and level sets of slope  $-q$ , the dashed lines in Figure 2; as  $q$  rises the supporting vertex rotates through the four in order of the rent they promise. The following lemma records this once, for a free scalar.

**Lemma 4** (Endpoint reduction). *Fix  $h > 0$  with  $\bar{\Delta}(h) > ch/M$  and let  $q \in \mathbb{R}$ . Define*

$$\begin{aligned} v_B(q) &= 0, & v_D(q) &= (1 - c)h + q r_D(h), \\ v_G(q) &= q M, & v_E(q) &= (1 - c)h + q r_E(h). \end{aligned} \quad (11)$$

Then

$$\max_{(r,s) \in \mathcal{A}(h)} \{s + qr\} = \max\{v_B(q), v_D(q), v_E(q), v_G(q)\},$$

and a maximizer may be chosen from  $\{A_B, A_D, A_E, A_G\}$ . Each is optimal on an interval of  $q$ , and

the intervals are arranged in order of rent: with

$$q_0(h) = -\frac{(1-c)h}{r_D(h)} < q_1 = 0 < q_2(h) = \frac{(1-c)h}{M - r_E(h)}, \quad (12)$$

the maximizer is  $A_B$  for  $q < q_0(h)$ ,  $A_D$  for  $q_0(h) < q < 0$ ,  $A_E$  for  $0 < q < q_2(h)$ , and  $A_G$  for  $q > q_2(h)$ , with ties at the breakpoints.

The four values in (11) are affine in  $q$  with slopes given by the rents (10), so a steeper action eventually overtakes a flatter one, any two cross at most once, and any switch as  $q$  rises is toward the action promising more rent. This is the source of the ordered structure of the optimal review menu.

The principal's own payoff,  $u = s - r$ , is the case  $q = -1$ . The lemma then leaves  $A_B$  and  $A_D$ , and  $A_D$  wins exactly when  $-1 > q_0(h)$ .

**Corollary 2** (The best continuation action). *The difficult standard  $A_D$  maximizes the principal's payoff over  $\mathcal{A}(h)$  if and only if*

$$(1-c)h \geq r_D(h), \quad \text{equivalently} \quad V^{\text{NR}}(h; c, M) \geq 0; \quad (13)$$

otherwise the bad stop  $A_B$  does.

Condition (13) is a restriction on  $h$ , not a property of the model. It holds at  $h = T$  by Assumption 5, and Figure 2 is drawn where it holds. But the discussion following Assumption 5 shows it *fails* for all sufficiently small  $h$ , since  $r_D(h)$  vanishes at rate  $\sqrt{h}$  up to a logarithmic factor while  $(1-c)h$  vanishes at rate  $h$ . With  $h = T - \tau$ , the content of Corollary 2 is that after a sufficiently late review no continuation contract is worth writing at all. This is the same force that makes very late reviews costly in Section 5, and it is why the bad stop is a genuine action rather than a formality.

Because the productivity of future effort does not depend on the review state, both stopping actions are purely incentive driven rather than responses to a good or bad project. The review problem of Section 4 is the problem of assigning these four actions to realizations of  $X_\tau$ .

## 4 Pre-review obedience and the incentive ladder

Having established what can be offered at the review, we now move backward to incentives before it, and ask which of the four actions the principal assigns to which review outcomes. Fix a review date  $\tau \in (0, T)$  and continue to write  $h = T - \tau$ , suppressing the dependence of  $r_D(h)$  and  $r_E(h)$  on  $h$  when no confusion can arise.

### 4.1 The fixed-review-date problem

The review state is  $X_\tau \sim N(\tau, \tau)$  if the agent worked throughout the first phase and  $X_\tau \sim N(0, \tau)$  if he shirked throughout; let  $g_1^\tau$  and  $g_0^\tau$  be the corresponding densities. Completing the square in

the two normal densities gives

$$\frac{g_1^\tau(x)}{g_0^\tau(x)} = \exp\left(x - \frac{\tau}{2}\right), \quad (14)$$

which is strictly increasing in  $x$ : the review signal satisfies the strict monotone likelihood ratio property. This is the only property of it used below, and it is what makes the optimal menu a monotone cutoff rule, since higher review states are stronger evidence of first-phase work.

Let  $a(x) = (r(x), s(x)) \in \mathcal{A}(h)$  be the action assigned after review state  $x$ , delivering the agent rent  $r(x)$  and the principal payoff  $s(x) - r(x)$ . Working through the first phase yields the agent  $\int r g_1^\tau - c\tau$  and shirking yields  $\int r g_0^\tau$ , so pre-review obedience requires

$$\int_{\mathbb{R}} r(x) [g_1^\tau(x) - g_0^\tau(x)] dx \geq c\tau. \quad (\text{OB}_\tau)$$

Because  $r(x) \geq 0$ , the utility from shirking is nonnegative, so  $(\text{OB}_\tau)$  also guarantees participation.

Conditional on inducing first-phase work, expected output accumulated by the review is  $\tau$ . The principal's fixed-review-date problem is therefore

$$\begin{aligned} V^R(\tau) = & \sup_{a(\cdot) = (r(\cdot), s(\cdot))} \tau + \int_{\mathbb{R}} [s(x) - r(x)] g_1^\tau(x) dx \\ & \text{subject to} \quad \int_{\mathbb{R}} r(x) [g_1^\tau(x) - g_0^\tau(x)] dx \geq c\tau, \\ & a(\cdot) \text{ measurable, } a(x) \in \mathcal{A}(h) \text{ for almost every } x. \end{aligned} \quad (\text{P}_\tau)$$

The principal chooses a continuation action separately for each review realization, but these choices are linked by the single constraint  $(\text{OB}_\tau)$ . Apart from it the problem is state by state, so a multiplier on that constraint separates the problem across review realizations. The value in  $(\text{P}_\tau)$  is conditional on inducing work in the first phase; inducing shirking, and dispensing with the review, are included when the review date is optimized below.

## 4.2 The Lagrangian for pre-review obedience

Let  $\lambda^* \geq 0$  denote a Lagrange multiplier associated with  $(\text{OB}_\tau)$  at the optimum, and define the *net shadow value of continuation rent*

$$Q_\lambda(x) \equiv \lambda \left(1 - \frac{g_0^\tau(x)}{g_1^\tau(x)}\right) - 1 = \lambda \left[1 - \exp\left(\frac{\tau}{2} - x\right)\right] - 1. \quad (15)$$

Writing the principal's payoff as  $s(x) - r(x)$ , adjoining  $(\text{OB}_\tau)$ , and factoring out  $g_1^\tau(x)$ , the Lagrangian is

$$\mathcal{L}(a; \lambda, \tau) = \tau - \lambda c\tau + \int_{\mathbb{R}} g_1^\tau(x) \{s(x) + Q_\lambda(x) r(x)\} dx, \quad (16)$$

and the usual optimality conditions require the optimal menu to maximize it at  $\lambda = \lambda^*$ , together with primal feasibility and complementary slackness. This is similar to the likelihood formulation of the principal's wage minimization problem familiar from agency theory (e.g., Holmström (1979)); here, however, the principal rewards favorable signals not by choosing a wage directly but by assigning continuation actions, and the surplus  $s(x)$  and the rent  $r(x)$  are linked by  $\mathcal{A}(h)$ .

The index  $Q_\lambda$  has a direct economic interpretation. Promising the agent one additional unit of continuation rent in review state  $x$  leaves total surplus unchanged and transfers one unit away from the principal, which is the  $-1$ . At the same time that unit contributes  $1 - g_0^\tau(x)/g_1^\tau(x)$  to the likelihood-weighted incentive for first-phase work, whose shadow value is  $\lambda^*$ . Corollary 2 is the case  $\lambda^* = 0$ , in which rent buys nothing and  $Q_{\lambda^*} \equiv -1$ .

The gain from introducing the multiplier is that  $(\text{OB}_\tau)$ , which links the rents assigned at all review realizations, has been replaced by a price  $\lambda^*$  for their incentive effects. The only remaining restriction is the pointwise one  $(r(x), s(x)) \in \mathcal{A}(h)$ , and the braces in (16) are exactly the objective of Lemma 4 at  $q = Q_{\lambda^*}(x)$ . The lemma therefore applies state by state: at every review realization the optimal action may be taken from  $\{A_B, A_D, A_E, A_G\}$ , and which one it is depends on the state only through the scalar  $Q_{\lambda^*}(x)$ .

### 4.3 The ordered review menu

Two features of the index turn that pointwise selection into an ordered menu. Whenever  $\lambda^* > 0$  it is strictly increasing,

$$Q'_{\lambda^*}(x) = \lambda^* \exp\left(\frac{\tau}{2} - x\right) > 0,$$

so better review outcomes strictly raise the net shadow value of rent; and by Lemma 4 the four actions are optimal on consecutive intervals of  $q$  ordered by the rent they promise. Composing the two, the optimal action can move only in the order

$$A_B \longrightarrow A_D \longrightarrow A_E \longrightarrow A_G,$$

switching at the states  $k_j^*$  at which  $Q_{\lambda^*}$  crosses  $q_j(h)$  from (12). In the knife-edge case in which a zero multiplier can support the optimum, an ordered menu can be selected from the principal's tied actions. Some upper regions may be empty in either case:  $A_j$  is used only if  $Q_{\lambda^*}$  ever reaches  $q_j(h)$ . Appendix B collects the cutoff formulas and the conditions determining which regions are active.

**Proposition 2** (At most four ordered review regions). *Fix  $\tau \in (0, T)$  and suppose first-phase work is implemented. Then an optimal review menu can be chosen to have at most four ordered performance regions:*

$$a^*(x) = \begin{cases} A_B, & x < k_0^*, \\ A_D, & k_0^* \leq x < k_1^*, \\ A_E, & k_1^* \leq x < k_2^*, \\ A_G, & x \geq k_2^*. \end{cases} \quad (17)$$

Whenever finite, the cutoffs satisfy  $k_0^* < k_1^* < k_2^*$ . An infinite cutoff means that the corresponding region and all regions above it are absent.

The bad stop assigns zero utility after sufficiently poor review performance. The difficult standard preserves second-phase production while providing the least costly positive reward for first-phase performance. Better review outcomes receive the easy standard, which sacrifices some second-phase profit in order to strengthen first-phase incentives. For exceptionally high review outcomes the principal may prefer the good stop: she pays the maximal prize  $M$  and abandons second-phase production because no work-inducing continuation contract can deliver as much utility to the agent. Both stop actions are inefficient; when used, they provide pre-review incentives—maximal punishment and maximal reward—at the cost of post-review continuation.

This ordering is an *incentive ladder*. When first-phase incentives are relatively easy to supply, the principal needs only the bad stop and the difficult standard. As the incentive requirement rises she first opens an easy-standard region and, if still more incentive power is needed, a top good-stop region. The richer three- and four-region menus therefore arise only when cheaper reward instruments are insufficient.

#### 4.4 Pre-review obedience and the multiplier

The multiplier and the menu are determined jointly. Proposition 2 characterizes maximization of the Lagrangian at a supporting multiplier; the remaining conditions are primal feasibility, nonnegativity, and complementary slackness. Let  $r^*(x)$  denote the continuation rent delivered by the optimal ordered menu, so that its pre-review incentive value is

$$\int_{k_0^*}^{k_1^*} r_D(h) [g_1^\tau - g_0^\tau] dx + \int_{k_1^*}^{k_2^*} r_E(h) [g_1^\tau - g_0^\tau] dx + \int_{k_2^*}^{\infty} M [g_1^\tau - g_0^\tau] dx,$$

with the obvious modifications when one or more upper regions are absent.

**Lemma 5** (Multiplier characterization). *An optimal solution to the fixed- $\tau$  problem conditional on inducing first-phase work can be chosen to have the ordered form in Proposition 2. A supporting multiplier  $\lambda^* \geq 0$  satisfies*

$$\lambda^* \left\{ \int_{\mathbb{R}} r^*(x) [g_1^\tau(x) - g_0^\tau(x)] dx - c\tau \right\} = 0.$$

Except in the knife-edge case  $(1 - c)h = r_D(h)$ , in which the principal can vary continuation rents without sacrificing post-review surplus, the supporting multiplier is strictly positive and pre-review obedience binds:

$$\int_{\mathbb{R}} r^*(x) [g_1^\tau(x) - g_0^\tau(x)] dx = c\tau. \quad (18)$$

The knife edge is exactly the boundary case of Corollary 2, at which  $q_0(h) = -1$  and the bad stop and the difficult standard are tied for the principal. Outside it, the cutoffs and the multiplier

are pinned down jointly in the usual way: the switching conditions determining  $k_0^*, k_1^*, k_2^*$  come from maximization of the Lagrangian at  $\lambda^*$ , while (18) requires the resulting menu to provide exactly the incentives needed for first-phase work. Appendix B gives the cutoff formulas and a more explicit characterization useful for computation.

## 5 Choosing the review date

We now allow the principal to choose the review date. For  $\tau \in (0, T)$ , continue to write

$$h = T - \tau.$$

Let  $a^*(\tau)$  denote an optimal fixed- $\tau$  menu and let  $\lambda^*$  be a supporting multiplier. By complementary slackness, the principal's optimized value satisfies

$$V^R(\tau) = \mathcal{L}(a^*(\tau); \lambda^*, \tau). \quad (19)$$

Hence, at any review date at which  $V^R$  is differentiable, the constrained envelope theorem gives its derivative from the direct effect of  $\tau$  on the Lagrangian.

To evaluate this derivative, it is convenient to put the distribution of the review signal on a fixed probability space. Conditional on first-phase work, write

$$X_\tau = \tau + \sqrt{\tau}Z, \quad Z \sim N(0, 1).$$

Substituting this expression into (15) gives

$$\tilde{Q}_\lambda(Z; \tau) \equiv Q_\lambda(\tau + \sqrt{\tau}Z) = \lambda \left[ 1 - \exp\left(-\frac{\tau}{2} - \sqrt{\tau}Z\right) \right] - 1. \quad (20)$$

The transformation is useful because  $Z$  has a distribution that does not depend on  $\tau$ . All effects of changing the review date therefore appear inside the integrand.

For a remaining horizon  $h$  and a net shadow value  $q$  of continuation rent, define

$$\mathcal{V}(q, h) \equiv \max \{0, (1 - c)h + qr_L(h), (1 - c)h + qr_H(h), qM\}. \quad (21)$$

By Lemma 4, this is exactly the maximized post-review contribution to the Lagrangian. The four terms correspond, respectively, to bad stop, low-rent continuation, high-rent continuation, and good stop.

Using (16), pointwise maximization over the post-review action, and complementary slackness, the optimized value can therefore be written as

$$V^R(\tau) = \tau - \lambda^* c\tau + \mathbb{E} \left[ \mathcal{V} \left( \tilde{Q}_{\lambda^*}(Z; \tau), T - \tau \right) \right]. \quad (22)$$

This equation is just the Lagrangian evaluated at the constrained optimum, written under the

standard-normal representation of the review signal.

### 5.1 First-Order Condition for Optimal Review

We now characterize how the principal's value varies with the review date. The constrained envelope theorem allows us to differentiate the optimized Lagrangian while holding fixed the supporting multiplier and, state by state, the post-review action selected by (21). Because the Gaussian review signal assigns zero probability to indifference points between actions, these switching points do not affect the derivative away from non-generic parameter values.

**Proposition 3** (Derivative of the Value Function). *Suppose the supporting multiplier  $\lambda^*$  is locally unique and the solution is away from a knife-edge parameter value. Then  $V^R$  is differentiable at  $\tau$  and*

$$\begin{aligned} \frac{dV^R(\tau)}{d\tau} = & 1 - \lambda^*c \\ & + \mathbb{E} \left[ \frac{\partial \mathcal{V}}{\partial q}(\tilde{Q}_{\lambda^*}(Z; \tau), h) \frac{\partial \tilde{Q}_{\lambda^*}(Z; \tau)}{\partial \tau} - \frac{\partial \mathcal{V}}{\partial h}(\tilde{Q}_{\lambda^*}(Z; \tau), h) \right], \end{aligned} \quad (23)$$

where  $h = T - \tau$ . Moreover, by the statewise envelope theorem,

$$\frac{\partial \mathcal{V}(q, h)}{\partial q} = R(q, h), \quad (24)$$

where  $R(q, h)$  is the rent delivered by the post-review action selected at  $(q, h)$ : it is 0,  $r_L(h)$ ,  $r_H(h)$ , or  $M$  according as the selected action is  $A_B$ ,  $A_L$ ,  $A_H$ , or  $A_G$ .

An interior optimal review date therefore satisfies

$$\frac{dV^R(\tau^*)}{d\tau} = 0. \quad (25)$$

At a regime transition, the corresponding condition uses the left and right derivatives.

The first two terms in (23) capture the direct effect of lengthening the first phase. Delaying the review increases first-phase production, giving the term 1, while also tightening the first-phase obedience requirement. The latter carries shadow value  $\lambda^*$  and raises the required incentive cost at rate  $c$ , giving the term  $-\lambda^*c$ .

The first term inside the expectation captures the information effect. Under work and shirking, the review-state means differ by  $\tau$  while their common standard deviation is  $\sqrt{\tau}$ . Their standardized separation is therefore  $\sqrt{\tau}$ , which increases with the review date. A later review makes the performance measure more informative about first-phase effort. This changes the review index  $\tilde{Q}_{\lambda^*}(z; \tau)$ , which is increasing in  $z$ . The value of that change is weighted by  $\partial \mathcal{V} / \partial q$ , the marginal value of increasing the review index. By (24), this marginal value is exactly the rent delivered by the action selected at that review state. Thus greater precision is especially valuable at histories at which the optimal policy assigns a large continuation rent.



The second term inside the expectation captures the opposing continuation-horizon effect. Delaying the review reduces  $h = T - \tau$  and therefore sacrifices  $\partial V / \partial h$  units of statewise continuation value. A longer continuation phase creates what may be called *incentive capital*: a family of work-inducing continuation contracts that generate real second-phase output while delivering different rents to the agent. By assigning these contracts according to the review outcome, the principal can recycle rents needed for second-phase incentives into rewards for first-phase effort. As  $h$  falls, the productive value of this continuation menu shrinks, and strong first-phase rewards increasingly require sacrificing second-phase production, culminating in the good-stop action  $A_G$ .

The optimal review date therefore does not merely trade an early noisy signal against a late precise signal. It balances the precision of the review against the productive contracting capacity that remains after the review. An early review preserves a valuable continuation phase but sorts the agent using a noisy measure of first-phase performance. A late review sorts performance more accurately but leaves less time in which continuation rents can be generated and recycled. An interior review is especially valuable when it allows the principal to use future agency rents as rewards for past effort without giving up too much future production. This mechanism is purely contractual: because the Brownian technology is translation invariant, the review state does not change the productivity of second-phase effort.

## 5.2 Endpoint behavior and the no-review benchmark

The no-review benchmark arises at either endpoint of the review-date problem. At  $\tau = 0$ , there is no first phase and the principal simply uses the minimum-rent full-effort contract over the entire horizon. At  $\tau = T$ , there is no second phase and the review itself is the terminal performance signal.

**Lemma 6** (Continuous extension to the endpoints). *Suppose Assumptions 4 and 5 hold. Then  $V^R$  is continuous on  $(0, T)$  and admits a continuous extension to  $[0, T]$  satisfying*

$$V^R(0) = V^R(T) = V^{\text{NR}}(T; c, M).$$

Consequently,

$$\sup_{\tau \in (0, T)} V^R(\tau) = \max_{\tau \in [0, T]} V^R(\tau) \geq V^{\text{NR}}(T; c, M). \quad (26)$$

The inequality in (26) reflects the principal's option to dispense with a genuine interior review. A review is strictly valuable if and only if

$$V^R(\tau) > V^{\text{NR}}(T; c, M)$$

for some  $\tau \in (0, T)$ .

It is useful to define the review gain

$$G(\tau) = V^R(\tau) - V^{\text{NR}}(T; c, M).$$

Lemma 6 gives

$$G(0) = G(T) = 0.$$

Thus a valuable review appears as a strictly positive interior hump in the review-gain function. If  $G(\tau) \leq 0$  throughout the interior, the principal optimally uses no review even though the fixed-review menu itself may have a rich cutoff structure.

### 5.3 One-sided endpoint derivatives

The endpoint equality in Lemma 6 makes one-sided derivatives especially informative. A negative right derivative at zero would favor an early review, while a negative left derivative at  $T$  implies that moving the review slightly to the left of the terminal date strictly raises the principal's payoff. Recalling the effects identified in the First-order condition (23), setting  $\tau = 0$  eliminates statistical precision while maximizing incentive capital, whereas setting  $\tau = T$  does the reverse. The two endpoints correspondingly behave very differently.

Recall from (AS) that  $\underline{M}(T, c)$  is the smallest cap that permits full effort without review. For  $M > \underline{M}(T, c)$ , let  $z_L(T)$  continue to denote the hard-threshold root of  $MD(T, z) = cT$ . Define

$$\ell_T = \frac{\phi(z_L(T) - \sqrt{T})}{\phi(z_L(T))} = \exp\left(z_L(T)\sqrt{T} - \frac{T}{2}\right) \in (0, 1) \quad (27)$$

The quantity  $\ell_T$  is the likelihood ratio of shirking to working at the payment threshold of the optimal no-review contract. It therefore measures the marginal incentive effectiveness of relaxing that threshold. In particular, the shadow value of the no-review incentive constraint is

$$\lambda_T^{\text{NR}} = \frac{1}{1 - \ell_T}.$$

This expression becomes especially informative as the payment cap approaches its feasibility bound. As  $M \downarrow \underline{M}(T, c)$ , the two incentive-compatible thresholds coalesce at  $T/2$ , so that  $z_L(T) \uparrow \sqrt{T}/2$  and  $\ell_T \uparrow 1$ . Consequently,  $\lambda_T^{\text{NR}} \rightarrow \infty$ . Thus, when the cap is nearly tight enough to make full effort infeasible, the no-review contract has essentially exhausted its incentive capacity: producing even a marginal increase in incentives has an arbitrarily large shadow cost.

**Proposition 4** (Endpoint derivatives and tight payment caps). *Suppose Assumptions 4 and 5 hold.*

(i) *The review value function has an infinitely negative right derivative at zero:*

$$\lim_{\tau \downarrow 0} \frac{V^R(\tau) - V^{\text{NR}}(T; c, M)}{\tau} = -\infty. \quad (28)$$

*Consequently, there exists  $\delta_0 > 0$  such that*

$$V^R(\tau) < V^{\text{NR}}(T; c, M) \quad \text{for every } \tau \in (0, \delta_0).$$

(ii) The left derivative at the terminal date equals the derivative of the no-review value with respect to its horizon:

$$V_-^{R'}(T) = \frac{\partial V^{\text{NR}}(T; c, M)}{\partial T} \quad (29)$$

$$= 1 - \lambda_T^{\text{NR}} c + \lambda_T^{\text{NR}} \frac{M\phi(z_L(T) - \sqrt{T})}{2\sqrt{T}} \quad (30)$$

$$= 1 - \frac{c - \frac{M\phi(z_L(T) - \sqrt{T})}{2\sqrt{T}}}{1 - \exp\left(z_L(T)\sqrt{T} - T/2\right)}. \quad (31)$$

In particular, a sufficient condition for a strictly valuable interior review is

$$c > 1 - \ell_T + \frac{M\phi(z_L(T) - \sqrt{T})}{2\sqrt{T}}. \quad (32)$$

(iii) The terminal derivative has the limiting behavior

$$\lim_{M \downarrow \underline{M}(T, c)} V_-^{R'}(T) = -\infty \quad (33)$$

and

$$\lim_{M \rightarrow \infty} V_-^{R'}(T) = 1 - c > 0. \quad (34)$$

The tight-cap conclusion can be stated directly in primitives.

**Corollary 3** (Tight caps make an interior review strictly valuable). *Suppose*

$$c \leq \frac{2\Phi(\sqrt{T}/2) - 1}{\Phi(\sqrt{T}/2)}. \quad (35)$$

*Equivalently, the no-review full-effort payoff is nonnegative at the feasibility boundary  $M = \underline{M}(T, c)$ . Then there exists  $\varepsilon > 0$  such that, for every*

$$M \in (\underline{M}(T, c), \underline{M}(T, c) + \varepsilon),$$

*the no-review full-effort contract is profitable and*

$$\sup_{\tau \in (0, T)} V^R(\tau) > V^{\text{NR}}(T; c, M) \geq 0.$$

*Thus, whenever the cap is sufficiently close to the smallest cap that can induce full effort, a genuine interior review is strictly optimal.*

Equation (28) has a distinct organizational interpretation. In this pure moral-hazard model, an arbitrarily short probationary phase *always* has negative incentive value. A very early review is too

noisy to distinguish work from shirking, yet bounded review rewards prevent the principal from magnifying its vanishing informational content at negligible cost. Any strictly optimal review must therefore be bounded away from date zero. Short probationary periods observed in practice must rest on motives absent from the model—for example, screening an unknown type, learning match quality, detecting technological failure, or preserving an option to abandon a bad project—rather than on the dynamic provision of effort incentives alone.

The derivative at  $\tau = T$  reveals the opposite margin. Moving the review a small distance to the left creates a short continuation phase. That phase has little direct productive value, but it creates a menu of continuation rents. These rents can be assigned according to first-phase performance and thereby serve as incentive prizes. When the payment cap is tight, the no-review contract is close to the statistical limit of what bounded rewards can implement: the hard and easy incentive-compatible thresholds nearly coincide, the shadow value  $\lambda_T^{\text{NR}}$  becomes arbitrarily large, and a small amount of continuation incentive capital has a first-order value. Proposition 7 shows further that the profitable reviews created in this terminal neighborhood use all four review regions. Corollary 3 then states the value conclusion directly in primitives: provided the no-review benchmark is nonnegative at the feasibility boundary, every sufficiently tight feasible cap makes an interior review strictly optimal.

With a loose cap, by contrast, the no-review contract can concentrate a large reward on a rare terminal outcome and induce effort with little rent. The shadow value of additional incentive instruments is then small, and (34) shows that the direct productive gain from lengthening the undivided project dominates near  $\tau = T$ . This does not rule out a more substantial interior review, but it eliminates the local terminal-date argument. The endpoint results thus expose two distinct requirements for a valuable review: the review must occur late enough to be informative, but early enough to leave a continuation phase capable of producing rents that can be recycled as incentives.

## 5.4 Review and Effort Feasibility

The same tight-cap region has an even stronger implication for implementability. At the no-review feasibility boundary, splitting the horizon creates strict incentive slack in both phases; indeed, review continues to permit full effort for caps slightly below the level required without review.

**Proposition 5** (A review Can restore feasibility). *Temporarily relax Assumption 4 and let  $M_0 \equiv \underline{M}(T, c)$ . At  $M = M_0$ , full effort without review is just feasible. For every genuine split  $\tau \in (0, T)$ , with  $h = T - \tau$ ,*

$$M_0 \overline{D}(\tau) > c\tau, \quad M_0 \overline{D}(h) > ch. \quad (36)$$

*Moreover, there exist a review date  $\tau_0 \in (0, T)$  and  $\varepsilon > 0$  such that, for every  $M \in (M_0 - \varepsilon, M_0)$ , full effort is infeasible without review, but a review at  $\tau_0$  admits a two-region menu that induces first-phase work and, whenever the project is continued, full continuation effort. Continuation occurs with positive probability.*

The economic implication is that a review can create incentives that are unavailable when performance is evaluated only at the terminal date. At the no-review feasibility boundary, the

payment cap has been pushed to its limit: even the most incentive-efficient terminal contract can just induce full effort. Introducing an interior review divides the project into two shorter incentive problems, each of which is strictly easier to implement. Formally, both phases satisfy the feasibility inequality strictly by (36).

More importantly, the review creates a new source of motivation. In the continuation phase there is now a nondegenerate range of work-inducing rents,  $r_L(h) < r_H(h)$ . The principal can condition these rents on the review outcome, promising more attractive continuation opportunities following outcomes that are more indicative of first-phase effort. Thus rents that must in any event be provided to motivate future effort can also serve as rewards for past effort. When still stronger incentives are required, the principal can supplement this continuation menu with stopping actions that deliver either zero or the maximal rent  $M$ .

## 5.5 Structure of the Incentive Ladder

\*\*\*Curt: Not sure this material is space-worthy. Maybe the first result? The proofs are currently located in the wrong place in the appendix...\*\*\*

It is useful to sharpen (23) by temporarily treating the lengths of the two phases as independent. Let  $\widehat{V}(\tau, h)$  denote the principal's optimized value when the first phase has length  $\tau$  and the continuation phase has length  $h$ , before imposing  $h = T - \tau$ . Along a fixed horizon,

$$\frac{dV^R(\tau)}{d\tau} = \widehat{V}_\tau(\tau, T - \tau) - \widehat{V}_h(\tau, T - \tau).$$

The simplest review menu turns out never to balance these two marginal effects.

**Proposition 6** (No smooth interior review date with two regions). *Suppose the fixed-date optimum is locally in the two-region regime  $A_B$ – $A_L$ , with a unique supporting multiplier and away from a regime boundary. Let  $\widehat{V}_2(\tau, h)$  denote the optimized value within this regime. Then*

$$G_2(\tau, h) \equiv \widehat{V}_{2,\tau}(\tau, h) - \widehat{V}_{2,h}(\tau, h) > 0. \quad (37)$$

*Consequently, after imposing  $h = T - \tau$ , the principal's value is strictly increasing in  $\tau$  along every smooth two-region segment. A smooth interior optimal review date therefore cannot lie in the two-region regime.*

Proposition 6 gives the timing tradeoff additional structure. Termination after bad outcomes and the minimum-rent continuation contract may be sufficient to induce first-phase work at a fixed review date, but they cannot support an interior choice of that date. At any smooth interior optimum the principal must also use the high-rent continuation action  $A_H$  on a nonempty set of review states. Thus the richer continuation ladder is not only a feature of the optimal fixed-date contract: it is what allows the marginal value of a longer first phase to be balanced against the marginal value of preserving continuation incentive capital.

The terminal derivative explored in the previous subsection also has a sharp implication for the form of the fixed-date menu near the terminal date. A three-region menu requires a supporting multiplier below the threshold at which the good-stop region appears. By contrast, when the terminal derivative is negative, the no-review incentive constraint has a shadow value large enough that this threshold must be crossed as the review date approaches  $T$ .

**Proposition 7** (Tight caps force the full incentive ladder near the terminal date). *Suppose the no-review full-effort benchmark is feasible and profitable. If*

$$V_-^{Rl}(T) < 0, \quad (38)$$

*then there exists  $\delta > 0$  such that the optimal fixed-review menu has all four regions*

$$A_B - A_L - A_H - A_G$$

*for every  $\tau \in (T - \delta, T)$ . Moreover, for some  $\tau \in (T - \delta, T)$ ,*

$$V^R(\tau) > V^{\text{NR}}(T; c, M).$$

*In particular, for all sufficiently tight feasible payment caps, a strictly valuable interior review exists and a strictly valuable review can be chosen with all four regions.*

Proposition 7 strengthens the endpoint result in two ways. A tight cap does not merely make a review valuable: the profitable reviews created near the terminal date use the entire incentive ladder. Bad stop provides the strongest punishment, the two continuation contracts recycle different levels of continuation rent, and good stop provides the strongest reward once the cap makes ordinary terminal incentives sufficiently scarce. Thus the good-stop region is not a curiosity of the fixed-date characterization; it is necessarily active in the terminal neighborhood that generates the tight-cap gain from review.

**Remark 1** (Three-region local optima). *Three-region local maxima can occur when the payment cap is sufficiently loose, so Proposition 6 does not extend mechanically from two regions to three. In extensive numerical searches, however, we have found no three-region local maximum when  $\lambda_T^{\text{NR}} > 1/c$ , and none under the stronger tight-cap condition  $V_-^{Rl}(T) < 0$ . In all such tight-cap cases examined, value is increasing through the three-region portion of the review-date domain and a local maximum occurs only after the good-stop region has entered.*

## 5.6 Computing and comparing the optimal review

There is no general ordering between an interior review and no review. The review can reduce agency rents by conditioning future opportunities on interim performance, but it may also destroy second-phase surplus through bad and good stops. Which effect dominates depends on  $(T, c, M)$ .

The following calculations solve the two continuation-root equations, the binding pre-review obedience equation, and the one-dimensional review-date problem.

For

$$T = 1, \quad c = 0.5, \quad M = 1.31,$$

the no-review value is

$$V^{\text{NR}} = 0.13379.$$

The optimal review is interior:

$$\tau^* = 0.47408, \quad V^R(\tau^*) = 0.20106.$$

Thus the review gain is approximately

$$0.06727.$$

The supporting multiplier is  $\lambda = 2.60063$ , and all four review regions are present. The standardized cutoffs are approximately

$$z_0 = 0.72305, \quad z_1 = -0.36064, \quad z_2 = -1.10160.$$

By contrast, for

$$T = 1, \quad c = 0.1, \quad M = 5,$$

the no-review value is

$$V^{\text{NR}} = 0.89387.$$

Numerical evaluation gives  $V^R(\tau) < V^{\text{NR}}$  for every interior review date. The supremum is approached as the review moves to an endpoint, so the principal optimally uses no genuine review.

The examples show that the endpoint result in Lemma 6 is substantive rather than merely technical. The optimized review may be strictly valuable, and when it is valuable the full four-region menu can occur. For other parameters, the rich fixed-review contract is dominated by the simpler no-review benchmark. The next natural step is to map the regions of  $(T, c, M)$  in which an interior review is valuable and to study how the optimal review date and the number of active review regions vary with the primitives.

## 6 Relation to the literature

This project combines several strands of the extant dynamic contracting literature: bounded compensation, interim performance evaluation, termination following poor performance, and the design or timing of feedback. We briefly review the relevant papers in each of these areas and discuss how our contribution builds on, and differs from, the existing approaches.

## 6.1 Bounded payments and likelihood-ratio bonuses

The static foundation is Jewitt et al. (2008), who study moral hazard with upper and lower bounds on compensation and show that optimal contracts place payments at the bounds over selected outcome regions. Under limited liability, their optimal contracts are naturally option-like. In a recent control setting, Jeong et al. (2026) derive likelihood-ratio-test payments when an agent chooses between controllers for a noisy linear dynamical system. These papers make clear that the upper-tail, zero-or- $M$  bonus used in our one-phase problem is not by itself the source of novelty. Our contribution instead begins with what happens when such bounded contracts are embedded in a genuinely dynamic review problem.

## 6.2 Interim evaluation, continuation, and termination

Several papers study interim performance evaluation with sequential effort and limited liability. Chen and Chiu (2013) analyze a two-stage project in which compensation and second-stage effort can depend on an interim evaluation. Lukas (2023) studies interim evaluation when outcomes across periods are interdependent. These papers establish that an interim signal can affect both incentives and continuation decisions, but their stages are fixed and their signals and technologies differ from the Gaussian cumulative-output environment considered here.

Termination following an unfavorable interim outcome also has important precedents. In Ray (2007), evaluation creates value by sorting projects and ending those with low early returns. Chao et al. (2014) study incentives around a stage-gate decision in a setting that combines information and effort problems, while Wong (2023) endogenizes dynamic monitoring and employment decisions. The termination motive in our model is deliberately different. Because Brownian output is additive and the productivity of future effort is independent of the review state, continuation has the same technological value after every realization of  $X_\tau$ . The bad stop is therefore purely incentive driven: it destroys otherwise valuable continuation opportunities in order to incentivise first-phase effort. At the other end of the performance distribution, the good stop pays  $M$  and forgoes second-phase effort because an extreme cash prize can be the cheapest way to supply sufficiently strong first-phase incentives.

## 6.3 Feedback design and review timing

The timing and disclosure of interim information have been studied in a variety of settings. Nafziger (2009) shows that the timing of information can have adverse incentive effects for a wealth-constrained agent. Ederer (2010) studies interim feedback in dynamic tournaments, where evaluation, motivation, and signal-jamming effects interact.

Most closely in spirit is the innovative article, Ely et al. (2025), who jointly design incentives and feedback in continuous time and obtain a silent phase followed by a transparent phase. Their principal controls what the agent learns about a coarse performance measure. If the agent *succeeds* before the end of the silent phase, the principal weights until the the end of the phase to tell him –



if he succeeds during the subsequent transparent phase, he is notified immediately. The principal’s objective in Ely et al. (2025) is to maximize the agent’s expected effort over the entire horizon, albeit without an explicit reason for doing so. By contrast, our principal receives the output produced by the agent net of compensation, which is the canonical agency objective. Here the review signal is public and Gaussian, and the principal chooses when to observe it and which bounded continuation opportunity to award in order to maximize this net-output payoff.

This distinction produces a different timing tradeoff. Waiting longer makes  $X_\tau$  more informative about first-phase effort, but it shortens the continuation phase and thereby reduces the stock of future rents available for incentive provision. The review date balances statistical precision against what may be called *incentive capital*. The endpoint results sharpen that tradeoff: an arbitrarily short probationary phase has negative incentive value, whereas a sufficiently tight payment cap makes an interior review strictly optimal whenever the no-review full-effort benchmark is nonnegative at the feasibility boundary.

## 6.4 Positioning of the present results

To our knowledge, the cited literature does not combine an optimally timed public Gaussian review, phase-level binary effort, limited liability, a hard payment cap, and an endogenous menu of stopping and continuation contracts. The distinctive results are the following. First, unavoidable continuation rents are recycled as prizes for first-phase effort. Second, strict MLRP and the linear continuation frontier generate an ordered menu with at most four regions: a bad stop, low-rent continuation, high-rent continuation, and a good stop. Third, both stopping regions are incentive instruments rather than responses to state-dependent continuation productivity. Finally, the review date is endogenous, and its endpoint derivatives yield sharp implications for probationary periods and for the role of tight versus loose payment caps.

The appropriate claim is therefore not that interim evaluation or threshold bonuses are new in isolation. Rather, the model identifies a new dynamic synthesis: future agency rents become scarce incentive resources, the principal allocates those resources through an ordered performance ladder, and the timing of the review determines both the quality of performance information and the quantity of rents available to reward it.

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# Appendix

## A Proofs for Section 3

In this section,  $h > 0$  is fixed and we suppress it from the notation where no confusion arises. It is convenient to normalize a contract by the cap and read it as a statistical test. For  $w : \mathbb{R} \rightarrow [0, M]$  put  $\varphi = w/M$ , so  $\varphi$  takes values in  $[0, 1]$ , and write

$$\alpha(\varphi) = \int \varphi f_0^h, \quad \beta(\varphi) = \int \varphi f_1^h \quad (\text{A.1})$$

for its size and power as a test of  $H_0 : e = 0$  against  $H_1 : e = 1$ . Then  $\mathbb{E}_0[w] = M\alpha(\varphi)$ ,  $\mathbb{E}_1[w] = M\beta(\varphi)$ , and  $(\text{IC}_h)$  reads  $\beta(\varphi) - \alpha(\varphi) \geq ch/M$ .

Two properties of the Gaussian family are used, and only these two. First, the likelihood ratio of (1) is

$$\ell^h(y) = \frac{f_0^h(y)}{f_1^h(y)} = \frac{\exp\{-y^2/2h\}}{\exp\{-(y-h)^2/2h\}} = \exp\left(\frac{h}{2} - y\right), \quad (\text{A.2})$$

which is finite, strictly positive and strictly decreasing — the monotone likelihood ratio property. Second,  $Y$  has a density, so no single outcome carries an atom.

### A.1 An exchange lemma

**Lemma A.1** (Neyman–Pearson, power-constrained form). *Let  $\varphi : \mathbb{R} \rightarrow [0, 1]$  be measurable, let  $k \in [-\infty, +\infty]$  satisfy  $\mathbb{P}\{Y \geq k \mid e = 1\} = \beta(\varphi)$ , and set  $\varphi^* = \mathbf{1}\{y \geq k\}$ . Then*

$$\beta(\varphi^*) = \beta(\varphi) \quad \text{and} \quad \alpha(\varphi^*) \leq \alpha(\varphi),$$

*with equality in the second only if  $\varphi = \varphi^*$  almost everywhere.*

*Proof.* The map  $k \mapsto \mathbb{P}\{Y \geq k \mid e = 1\}$  is a continuous, strictly decreasing bijection onto  $[0, 1]$ , so  $k$  exists, is unique, and  $\beta(\varphi^*) = \beta(\varphi)$  by construction. Hence  $\int (\varphi - \varphi^*) f_1^h = 0$ . Writing  $\ell_k^h = \ell^h(k)$  and subtracting  $\ell_k^h$  times that vanishing integral,

$$\alpha(\varphi) - \alpha(\varphi^*) = \int (\varphi - \varphi^*) \ell^h f_1^h = \int (\varphi - \varphi^*) [\ell^h - \ell_k^h] f_1^h. \quad (\text{A.3})$$

For  $y > k$  we have  $\varphi^* = 1 \geq \varphi$  and, by (A.2),  $\ell^h(y) < \ell_k^h$ : the integrand is a product of two nonpositive factors. For  $y < k$  we have  $\varphi^* = 0 \leq \varphi$  and  $\ell^h(y) > \ell_k^h$ : a product of two nonnegative factors. The set  $\{y = k\}$  is null. The integrand in (A.3) is therefore nonnegative almost everywhere, which gives  $\alpha(\varphi^*) \leq \alpha(\varphi)$ . If the integral vanishes then, since  $f_1^h > 0$  and  $\ell^h(y) \neq \ell_k^h$  for every  $y \neq k$ , we must have  $\varphi = \varphi^*$  almost everywhere.  $\square$

Because  $Y$  has a density, the extremal test is a pure indicator and no randomization on  $\{\ell^h = \ell_k^h\}$

is required. This is why the extreme contracts of Proposition 1 are indicators rather than schedules that are interior on a set of positive measure.

## A.2 Proof of Lemma 1

*Proof of Lemma 1.* Put  $\varphi = w/M$  and let  $k$  solve  $\mathbb{P}\{Y \geq k \mid e = 1\} = \beta(\varphi)$ , so that  $\tilde{w} = M\mathbf{1}\{y \geq k\}$  is exactly the contract of the statement: it satisfies  $\mathbb{E}_1[\tilde{w}] = M\beta(\varphi) = \mathbb{E}_1[w]$ , and it is the unique threshold contract that does so. Lemma A.1 gives  $\alpha(\varphi^*) \leq \alpha(\varphi)$ , that is  $\mathbb{E}_0[\tilde{w}] \leq \mathbb{E}_0[w]$ , with equality only if  $w = \tilde{w}$  almost everywhere. Subtracting from  $\mathbb{E}_1[\tilde{w}] = \mathbb{E}_1[w]$ ,

$$\mathbb{E}_1[\tilde{w}] - \mathbb{E}_0[\tilde{w}] \geq \mathbb{E}_1[w] - \mathbb{E}_0[w],$$

so if  $w$  satisfies  $(IC_h)$  then so does  $\tilde{w}$ . □

## A.3 Proof of Lemma 2

*Proof of Lemma 2.* (i) For the threshold test  $\varphi^* = \mathbf{1}\{y \geq k\}$  and the normalized bar  $z = (h-k)/\sqrt{h}$ , standardizing and using  $-k/\sqrt{h} = z - \sqrt{h}$  gives

$$\beta(\varphi^*) = \Phi(z), \quad \alpha(\varphi^*) = \Phi(z - \sqrt{h}), \quad \beta - \alpha = D(h, z),$$

which is (2). Since  $\partial_z D(h, z) = \phi(z) - \phi(z - \sqrt{h})$  vanishes only at  $z = \sqrt{h}/2$  and changes sign there from positive to negative,  $D(h, \cdot)$  is strictly unimodal with maximum  $\bar{D}(h) = 2\Phi(\sqrt{h}/2) - 1$ , and  $D(h, z) \rightarrow 0$  as  $z \rightarrow \pm\infty$ . By Lemma A.1 every  $\varphi$  is weakly dominated, in the sense of a weakly larger spread  $\beta - \alpha$ , by the threshold test of the same power, so

$$\max_{0 \leq w \leq M} [\mathbb{E}_1[w] - \mathbb{E}_0[w]] = M \max_z D(h, z) = M \bar{D}(h).$$

Comparing with  $(IC_h)$ ,  $\mathcal{F}_h \neq \emptyset$  exactly when  $\bar{D}(h) \geq ch/M$ .

(ii) Put  $a = \sqrt{h}/2$ , a strictly increasing bijection of  $(0, \infty)$  onto itself, so that  $h = 4a^2$  and

$$\frac{\bar{D}(h)}{h} = G(a) \equiv \frac{2\Phi(a) - 1}{4a^2}.$$

It suffices that  $G$  be strictly decreasing. Differentiating,

$$G'(a) = \frac{a\phi(a) - [2\Phi(a) - 1]}{2a^3} = -\frac{F(a)}{2a^3}, \quad F(a) = 2\Phi(a) - 1 - a\phi(a),$$

and  $F(0) = 0$  with  $F'(a) = 2\phi(a) - \phi(a) - a\phi'(a) = (1 + a^2)\phi(a) > 0$ , using  $\phi'(a) = -a\phi(a)$ . Hence  $F > 0$  on  $(0, \infty)$  and  $G' < 0$ . If  $\bar{D}(T)/T > c/M$  and  $h \in (0, T]$ , monotonicity gives  $\bar{D}(h)/h \geq \bar{D}(T)/T > c/M$ .

(iii) By Lemma 1 a threshold contract with bar  $z$  lies in  $\mathcal{F}_h$  if and only if  $D(h, z) \geq ch/M$ . By the strict unimodality established in (i), together with  $D(h, z) \rightarrow 0$  at  $\pm\infty$  and  $\bar{D}(h) > ch/M$ , the

equation  $D(h, z) = ch/M$  has exactly two roots  $z_L(h) < \sqrt{h}/2 < z_H(h)$ , and the superlevel set  $\{z : D(h, z) \geq ch/M\}$  is precisely the closed interval  $[z_L(h), z_H(h)]$ , with the inequality strict on its interior.  $\square$

#### A.4 Proof of Proposition 1

*Proof of Proposition 1.* By (4) the rent  $r(w; h) = \mathbb{E}_1[w] - ch$  is an increasing affine function of  $\mathbb{E}_1[w]$ , so it suffices to extremize  $\mathbb{E}_1[w]$  over  $\mathcal{F}_h$ . The two endpoints are handled by the same argument.

*Both extremes are attained at the two extreme bars.* Both  $w_L^h$  and  $w_H^h$  belong to  $\mathcal{F}_h$ , since  $\Delta(h, z_j(h)) = ch/M$  for  $j \in \{D, E\}$  by definition of the roots. Now take any  $w \in \mathcal{F}_h$  and let  $\tilde{w} = M\mathbf{1}\{y \geq k\}$  be the threshold contract with  $\mathbb{E}_1[\tilde{w}] = \mathbb{E}_1[w]$  supplied by Lemma 1. That lemma gives  $\tilde{w} \in \mathcal{F}_h$ , so its normalized bar  $z = (h - k)/\sqrt{h}$  satisfies  $D(h, z) \geq ch/M$ , and by Lemma 2(iii)  $z \in [z_L(h), z_H(h)]$ . Since  $\mathbb{E}_1[\tilde{w}] = M\Phi(z)$  with  $\Phi$  strictly increasing,

$$M\Phi(z_L(h)) = \mathbb{E}_1[w_L^h] \leq \mathbb{E}_1[w] \leq \mathbb{E}_1[w_H^h] = M\Phi(z_H(h)). \quad (\text{A.4})$$

These bounds hold over all of  $\mathcal{F}_h$  and not merely over threshold contracts, so the minimum and the maximum are attained at  $w_L^h$  and  $w_H^h$  respectively.

*Uniqueness and binding.* Suppose  $\mathbb{E}_1[w] = \mathbb{E}_1[w_j^h]$  for  $j \in \{D, E\}$ . Then  $z = z_j(h)$  in the display above, so  $\tilde{w} = w_j^h$ , and since the constraint binds at that bar,

$$\mathbb{E}_0[\tilde{w}] = \mathbb{E}_1[\tilde{w}] - M\Delta(h, z_j(h)) = \mathbb{E}_1[w] - ch.$$

On the other hand  $w \in \mathcal{F}_h$  gives  $\mathbb{E}_0[w] \leq \mathbb{E}_1[w] - ch = \mathbb{E}_0[\tilde{w}]$ , while Lemma A.1 gives the reverse inequality  $\mathbb{E}_0[\tilde{w}] \leq \mathbb{E}_0[w]$ . Hence  $\mathbb{E}_0[w] = \mathbb{E}_0[\tilde{w}]$ , and the equality case of that lemma yields  $w = w_j^h$  almost everywhere. The same computation shows that  $(\text{IC}_h)$  binds at  $w_L^h$  and at  $w_H^h$ . This proves (i).

*Part (ii).* Because  $(\text{IC}_h)$  binds,  $\mathbb{E}_1[w_j^h] - \mathbb{E}_0[w_j^h] = ch$ , so

$$r_j(h) = \mathbb{E}_1[w_j^h] - ch = \mathbb{E}_0[w_j^h] = M\Phi(z_j(h) - \sqrt{h}),$$

which is (5). Positivity and  $r_L(h) < r_H(h)$  follow from strict monotonicity of  $\Phi$  together with  $z_L(h) < z_H(h)$ , and  $r_H(h) = M\Phi(z_H(h)) - ch < M - ch$  because  $\Phi(z_H(h)) < 1$ .

*Part (iii).* The set  $\mathcal{F}_h$  is convex, being defined by the single linear inequality  $(\text{IC}_h)$  on the convex set  $\{0 \leq w \leq M\}$ , and  $w \mapsto r(w; h)$  is affine; hence  $\mathcal{R}_h$  is an interval. By (i) and (ii) its endpoints are  $r_L(h)$  and  $r_H(h)$  and both are attained, so  $\mathcal{R}_h = [r_L(h), r_H(h)]$ . Explicitly, the mixture  $w_\alpha^h = (1 - \alpha)w_L^h + \alpha w_H^h$  lies in  $\mathcal{F}_h$  with the constraint binding and delivers rent  $(1 - \alpha)r_L(h) + \alpha r_H(h)$ . Substituting into (4) gives (6).  $\square$

#### A.5 Proof of Corollary 1

*Proof of Corollary 1.* By Proposition 1(iii) the principal's surplus over  $\mathcal{F}_h$  is  $S_h(r) = (1 - c)h - r$  with  $r$  ranging over  $[r_L(h), r_H(h)]$ , and  $S_h$  is strictly decreasing. Its maximum is therefore attained

at  $r = r_L(h)$ , which by Proposition 1(i) is delivered by  $w_L^h$  and, up to a null set, only by  $w_L^h$ . Substituting  $r_L(h) = M\Phi(z_L(h)) - ch$  from (5),

$$s_L(h) = (1 - c)h - r_L(h) = h - M\Phi(z_L(h)),$$

which is (7). □

## A.6 Proof of Lemma 3

*Proof of Lemma 3.* Write

$$p_1(z) = \Phi(z), \quad p_0(z) = \Phi(z - \sqrt{h}), \quad q(z) = \frac{p_0(z)}{p_1(z)},$$

so that  $p_1 - p_0 = D(h, \cdot)$  and, by Proposition 1(i), the difficult bar satisfies the binding condition

$$M[p_1(z_L(h)) - p_0(z_L(h))] = ch, \quad \text{equivalently} \quad \Delta(h, z_L(h)) = \frac{ch}{M}. \quad (\text{A.5})$$

*The bar falls as the cap rises.* The right-hand side of (A.5) is strictly decreasing in  $M$ . By Lemma 2(iii) the root  $z_L(h)$  lies on the branch  $(-\infty, \sqrt{h}/2)$ , on which  $D(h, \cdot)$  is strictly increasing. Hence  $z_L(h)$  is strictly decreasing in  $M$ : relaxing the cap lets the principal push the reward farther into the upper tail.

*The tail ratio.* The normal inverse Mills ratio  $x \mapsto \phi(x)/\Phi(x)$  is strictly decreasing, so for every  $z$ ,

$$\frac{d}{dz} \log q(z) = \frac{\phi(z - \sqrt{h})}{\Phi(z - \sqrt{h})} - \frac{\phi(z)}{\Phi(z)} > 0.$$

Thus  $q$  is strictly increasing, and combining with the previous paragraph,  $q(z_L(h))$  is strictly decreasing in  $M$ .

*Monotonicity of the rent.* Dividing (A.5) by  $p_1(z_L(h))$  and rearranging,

$$M p_1(z_L(h)) = \frac{ch}{1 - q(z_L(h))}. \quad (\text{A.6})$$

The denominator is strictly increasing in  $M$ , so expected compensation  $M\Phi(z_L(h))$  is strictly decreasing in  $M$ . Since the effort cost  $ch$  is fixed,  $R^{\text{NR}}(h; c, M) = M\Phi(z_L(h)) - ch$  is strictly decreasing in  $M$  as well.

*The limit.* As  $M \rightarrow \infty$ , (A.5) forces  $\Delta(h, z_L(h)) \rightarrow 0$  along the increasing branch, so  $z_L(h) \rightarrow -\infty$ . By the Mills ratio asymptotic  $\Phi(x) \sim \phi(x)/|x|$  as  $x \rightarrow -\infty$ ,

$$q(z_L(h)) = \frac{\Phi(z_L(h) - \sqrt{h})}{\Phi(z_L(h))} \sim \frac{|z_L(h)|}{|z_L(h) - \sqrt{h}|} \exp\left(z_L(h)\sqrt{h} - \frac{h}{2}\right) \rightarrow 0.$$

By (A.6),  $M\Phi(z_L(h)) \downarrow ch$  and therefore  $R^{\text{NR}}(h; c, M) \rightarrow 0$ . □

## A.7 Proof of Lemma 4

*Proof of Lemma 4.* On  $\mathcal{A}_0(h)$  we have  $s = 0$  and  $r \in [0, M]$ , so the objective  $s + qr$  equals  $qr$ ; it is maximized at  $r = 0$  if  $q < 0$  and at  $r = M$  if  $q > 0$ , that is at  $A_B$  and at  $A_G$  respectively, every point being tied when  $q = 0$ . On  $\mathcal{A}_1(h)$  we have  $s = (1 - c)h$  and  $r \in [r_D(h), r_E(h)]$ , so the objective equals  $(1 - c)h + qr$ ; it is maximized at  $r_D(h)$  if  $q < 0$  and at  $r_E(h)$  if  $q > 0$ , that is at  $A_D$  and at  $A_E$ , again with ties when  $q = 0$ . A maximizer over  $\mathcal{A}(h) = \mathcal{A}_0(h) \cup \mathcal{A}_1(h)$  may therefore be chosen from  $\{A_B, A_D, A_E, A_G\}$ , and the values these four deliver are exactly  $v_B, v_D, v_E, v_G$  in (11).

It remains to rank them. For  $q < 0$ ,

$$v_D(q) - v_E(q) = q[r_D(h) - r_E(h)] > 0 \quad \text{and} \quad v_B(q) = 0 > qM = v_G(q),$$

so only  $A_B$  and  $A_D$  can be optimal, and  $v_D(q) \geq v_B(q)$  if and only if  $q \geq -(1 - c)h/r_D(h) = q_0(h)$ . For  $q > 0$  the first inequality reverses, so  $v_E(q) > v_D(q)$ , and  $v_E(q) > 0 = v_B(q)$ ; the remaining comparison is

$$v_E(q) - v_G(q) = (1 - c)h - q[M - r_E(h)],$$

which is positive exactly when  $q < (1 - c)h/[M - r_E(h)] = q_2(h)$ . At  $q = 0$ ,  $v_D(0) = v_E(0) = (1 - c)h > 0 = v_B(0) = v_G(0)$ . This gives the four regions of (12), with the two adjacent actions tied at each breakpoint.  $\square$

## A.8 Two remarks used in the text

The following two facts are asserted in Section 3 and are recorded here for completeness.

**Lemma A.2** (Short phases are unprofitable).  $V^{\text{NR}}(h; c, M) = h - M\Phi(z_L(h)) < 0$  for all sufficiently small  $h > 0$ .

*Proof.* Feasibility on  $(0, T]$  is Lemma 2(ii), so  $z_L(h)$  is well defined. Since  $\Delta(h, z_L(h)) = ch/M \rightarrow 0$  and  $z_L(h) < \sqrt{h}/2$ , we have  $z_L(h) \rightarrow -\infty$  as  $h \downarrow 0$ . Write  $u = \sqrt{h}$  and  $z = z_L(h)$ . From  $\Phi(z) - \Phi(z - u) = cu^2/M$  and the mean value theorem there is  $\xi \in (z - u, z)$  with  $u\phi(\xi) = cu^2/M$ , that is  $\phi(\xi) = cu/M$ . Hence  $\xi \rightarrow -\infty$  with  $\xi^2/2 \sim \log(1/u)$ , so  $|z| = O(\sqrt{\log(1/u)})$  and, by the Mills ratio bound  $\Phi(z) \geq \phi(z)/(|z| + 1)$  together with  $\phi(z) \geq \phi(\xi) = cu/M$ ,

$$M\Phi(z_L(h)) \geq \frac{M\phi(z)}{|z| + 1} \geq \frac{cu}{|z| + 1} = \Omega\left(\frac{\sqrt{h}}{\sqrt{\log(1/h)}}\right).$$

Expected compensation therefore vanishes at rate  $\sqrt{h}$  up to a logarithmic factor, while expected output vanishes at rate  $h$ . Hence  $h - M\Phi(z_L(h)) < 0$  for all sufficiently small  $h$ .  $\square$

**Lemma A.3** (Extreme points of the review action set). Fix  $h$  with  $\bar{D}(h) > ch/M$  and let  $\mathcal{A}(h) = \mathcal{A}_0(h) \cup \mathcal{A}_1(h)$  be as in (9). For every  $q \in \mathbb{R}$  a maximizer of  $s + qr$  over  $\mathcal{A}(h)$  can be chosen from  $\{A_B, A_L, A_H, A_G\}$ .

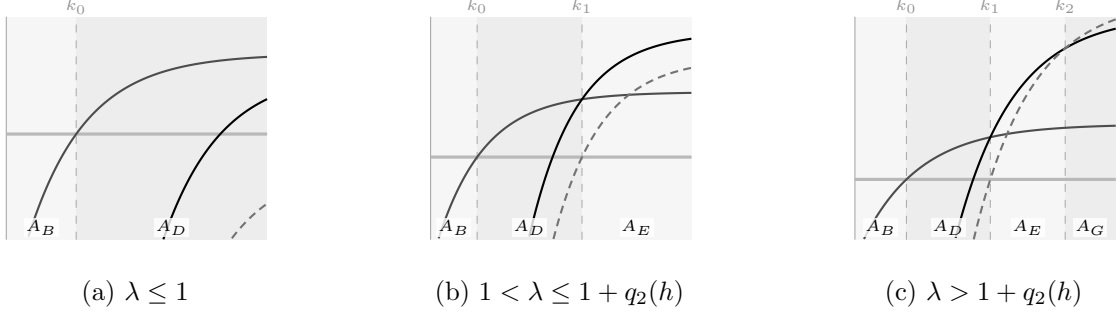


Figure 3: The four values  $v_j(Q_\lambda(x))$  against the review state  $x$ , for  $\tau = h = 1$ ,  $c = 0.3$ ,  $M = 1.2$  and  $\lambda = 0.8, 2.0, 3.5$ . The flat line is  $v_B$ ; the two solid curves are  $v_D$  (flatter) and  $v_E$ ; the dashed curve is  $v_G$ . Each is affine in  $Q_\lambda(x)$  with slope equal to the rent the action promises, and the optimal action attains the upper envelope, so the actions are used in order of the rent they carry. Raising  $\lambda$  raises the ceiling  $\lim_{x \rightarrow \infty} Q_\lambda(x) = \lambda - 1$  and brings the upper regions into play. Vertical scales differ across panels.

*Proof.* Each of  $\mathcal{A}_0(h)$  and  $\mathcal{A}_1(h)$  is a compact line segment, with endpoints  $\{A_B, A_G\}$  and  $\{A_L, A_H\}$  respectively. The objective  $s + qr$  is affine in  $(r, s)$ , so on each segment it attains its maximum at an endpoint; taking the better of the two segment maxima gives a maximizer over the union. That  $\mathcal{A}_1(h)$  has the stated endpoints is Proposition 1(iii); that  $\mathcal{A}_0(h)$  does is immediate, since a shirking agent promised  $r \in [0, M]$  can be paid the constant  $w \equiv r$ , which is feasible exactly when  $r \leq M$ .  $\square$

## B Proofs for Section 4

### B.1 Cutoffs and active regions

Lemma 4 ranks the four actions by the scalar  $q$ , with breakpoints (12). Two features of the index  $Q_\lambda$  turn that ranking into a menu. It is strictly increasing in  $x$ , so the regions are intervals of review states arranged in the same order; and it is a bijection from  $\mathbb{R}$  onto  $(-\infty, \lambda - 1)$ , since

$$\lim_{x \rightarrow \infty} Q_\lambda(x) = \lambda - 1. \quad (\text{A.7})$$

The action  $A_j$  is therefore used on a nonempty region exactly when the index can reach  $q_j$ , that is when  $\lambda - 1 > q_j$ , and the cutoff is the state at which it does so.

For  $j \in \{0, 1, 2\}$ , the corresponding cutoff is therefore

$$k_j(\lambda) = \frac{\tau}{2} - \log \left( 1 - \frac{1 + q_j(h)}{\lambda} \right) \quad \text{if } \lambda > 1 + q_j(h), \quad k_j(\lambda) = +\infty \quad \text{otherwise.} \quad (\text{A.8})$$

Whenever finite, the cutoffs are strictly ordered. If one cutoff is infinite, that region and every higher-rent region are absent.

Two consequences are worth recording. First, the bad-stop cutoff is finite precisely when



$\lambda > 1 + q_0(h)$ , which is why

$$\underline{\lambda}(h) = \max \{0, 1 + q_0(h)\} = \max \left\{ 0, 1 - \frac{(1-c)h}{r_D(h)} \right\}$$

is the smallest multiplier that can support first-phase work: below it the selected review-state rent is constant across states and hence supplies no incentives for first-phase work. Second, because the breakpoints are ordered, the regions enter one at a time as  $\lambda$  rises,

| Multiplier range                          | Review regions           |
|-------------------------------------------|--------------------------|
| $\underline{\lambda}(h) < \lambda \leq 1$ | $A_B - A_D,$             |
| $1 < \lambda \leq 1 + q_2(h)$             | $A_B - A_D - A_E,$       |
| $\lambda > 1 + q_2(h)$                    | $A_B - A_D - A_E - A_G,$ |

which is the content of the three panels of Figure 3.

For each cutoff, define its standardized value under first-phase work by  $z_j(\lambda, \tau) = [\tau - k_j(\lambda)] / \sqrt{\tau}$ . Substituting (A.8) gives the closed form

$$z_j(\lambda, \tau) = \frac{\sqrt{\tau}}{2} + \frac{1}{\sqrt{\tau}} \log \left( 1 - \frac{1 + q_j(h)}{\lambda} \right), \quad j \in \{0, 1, 2\}, \quad (\text{A.9})$$

with the convention  $z_j(\lambda, \tau) = -\infty$  when  $k_j(\lambda) = +\infty$ . Recall from (3) that  $\Delta(\tau, z) = \Phi(z) - \Phi(z - \sqrt{\tau})$  is the increase in the probability of crossing the associated upper-tail cutoff when the agent works rather than shirks in the first phase.

Under the menu of Proposition 2 the agent's review-state utility is the increasing step function

$$r_\lambda(x) = r_D(h) \mathbf{1}\{x \geq k_0(\lambda)\} + [r_E(h) - r_D(h)] \mathbf{1}\{x \geq k_1(\lambda)\} + [M - r_E(h)] \mathbf{1}\{x \geq k_2(\lambda)\},$$

so the incentive it supplies is

$$\begin{aligned} I(\lambda; \tau) &= r_D(h) \Delta(\tau, z_0(\lambda, \tau)) \\ &\quad + [r_E(h) - r_D(h)] \Delta(\tau, z_1(\lambda, \tau)) \\ &\quad + [M - r_E(h)] \Delta(\tau, z_2(\lambda, \tau)). \end{aligned} \quad (\text{A.10})$$

The convention  $\Delta(\tau, -\infty) = 0$  automatically removes terms corresponding to absent regions.

**Lemma A.4** (Monotonicity of incentive supply). *The incentive supply  $I(\lambda; \tau)$  is weakly increasing in  $\lambda$ . It is strictly increasing whenever a change in  $\lambda$  changes the selected review-state utility on a set of positive probability away from indifference points.*

**Remark 2** (Existence and possible nonuniqueness of the multiplier). *Outside the knife-edge case  $(1-c)h = r_D(h)$ , the cutoff formulas imply that  $I(\lambda; \tau)$  is continuous on the relevant multiplier range. It approaches zero as  $\lambda$  decreases to  $\underline{\lambda}(h)$ , while Lemma A.4 shows that it is weakly increasing.*

As  $\lambda \rightarrow \infty$  the menu converges to a payment of  $M$  exactly when  $X_\tau \geq \tau/2$ , and hence

$$I(\lambda; \tau) \longrightarrow M \left[ 2\Phi \left( \frac{\sqrt{\tau}}{2} \right) - 1 \right] = M \bar{\Delta}(\tau).$$

Since  $\tau < T$ , Assumption 4 and Lemma 2(ii) give  $\bar{\Delta}(\tau)/\tau > c/M$ , so this limit exceeds  $c\tau$  and a solution to (18) exists. The solution is generically unique. If  $(1-c)h = r_D(h)$ , the bad stop and the difficult standard are tied at  $\lambda = 0$ ; mixing between tied actions supplies any required incentive up to the corresponding threshold, so existence still obtains, although the supporting multiplier need not be unique.

## B.2 When the upper review regions become active

Proposition 2 describes the menu through the supporting multiplier. Monotonicity of incentive supply converts this into a statement about the primitive incentive requirement  $c\tau$ , because the multiplier is pinned down by  $I(\lambda; \tau) = c\tau$ . By (A.8) the region using  $A_E$  appears as  $\lambda$  crosses

$$\lambda_2 = 1 + q_1 = 1, \quad \text{and the region using } A_G \text{ appears as } \lambda \text{ crosses} \quad \lambda_3 = 1 + q_2(h), \quad (\text{A.11})$$

so the incentive levels at which the menu gains a region are the supplies at these two multipliers.

**Proposition A.1** (Boundaries between the two-, three-, and four-region menus). *Fix  $\tau \in (0, T)$  and let*

$$\mathcal{I}_2(\tau) = I(\lambda_2; \tau), \quad \mathcal{I}_3(\tau) = I(\lambda_3; \tau), \quad (\text{A.12})$$

*evaluated from (A.10) and (A.9). Absent regions drop out through the convention  $\Delta(\tau, -\infty) = 0$ , so explicitly*

$$\mathcal{I}_2(\tau) = r_D(h) \Delta(\tau, z_0(\lambda_2, \tau)), \quad \mathcal{I}_3(\tau) = \mathcal{I}_2^{\lambda_3}(\tau) + [r_E(h) - r_D(h)] \Delta(\tau, z_1(\lambda_3, \tau)),$$

*where  $\mathcal{I}_2^{\lambda_3}(\tau) = r_D(h) \Delta(\tau, z_0(\lambda_3, \tau))$ . Then  $\mathcal{I}_2(\tau) \leq \mathcal{I}_3(\tau)$ , with strict inequality generically, and the optimal fixed- $\tau$  menu is*

| <i>First-phase incentive requirement</i>               | <i>Optimal review regions</i> |        |
|--------------------------------------------------------|-------------------------------|--------|
| $c\tau \leq \mathcal{I}_2(\tau)$                       | $A_B - A_D,$                  |        |
| $\mathcal{I}_2(\tau) < c\tau \leq \mathcal{I}_3(\tau)$ | $A_B - A_D - A_E,$            |        |
| $c\tau > \mathcal{I}_3(\tau)$                          | $A_B - A_D - A_E - A_G.$      | (A.13) |

*At either equality the newly appearing region has zero probability.*

**Remark 3** (Interpretation of the region boundaries). *The two boundaries order the principal's reward instruments by cost. Below  $\mathcal{I}_2(\tau)$  the cheap instruments suffice: termination after poor outcomes and the difficult standard otherwise. Between the boundaries she must also promise the easy standard after good outcomes, sacrificing second-phase profit. Above  $\mathcal{I}_3(\tau)$  even the full range*

of continuation rents is insufficient at the margin, and she opens a top region paying  $M$  and forgoing second-phase production altogether.

### B.3 The fixed-review value

Let  $u_D(h) = (1 - c)h - r_D(h)$  and  $u_E(h) = (1 - c)h - r_E(h)$  be the principal's payoffs under the two continuation standards. Conditional on first-phase work, the probabilities of the four regions are determined by the standardized cutoffs, and the principal's value from the optimal fixed- $\tau$  menu is

$$\begin{aligned} V^R(\tau) = & \tau + u_D(h) [\Phi(z_0(\lambda, \tau)) - \Phi(z_1(\lambda, \tau))] \\ & + u_E(h) [\Phi(z_1(\lambda, \tau)) - \Phi(z_2(\lambda, \tau))] - M\Phi(z_2(\lambda, \tau)), \end{aligned} \quad (\text{A.14})$$

where  $\lambda$  satisfies (18) and  $\Phi(-\infty) = 0$ . Equation (A.14) applies to the two-, three-, and four-region cases under the cutoff conventions above.

### B.4 Remaining proofs

*Proof of Proposition 2.* By Lemma 4 a pointwise maximizer of the braces in (16) may be selected from  $\{A_B, A_D, A_E, A_G\}$ , ranked by  $q$  according to the breakpoints (12). Substituting  $q = Q_\lambda(x)$  and using strict monotonicity of  $Q_\lambda$  turns that ranking into the ordered rule (17). Since  $Q_\lambda$  is a strictly increasing bijection onto  $(-\infty, \lambda - 1)$ , the equation  $Q_\lambda(k) = q_j(h)$  has a solution, necessarily unique, if and only if  $q_j(h) < \lambda - 1$ , and inverting gives (A.8); otherwise  $Q_\lambda$  never reaches  $q_j(h)$  and the region is empty. Finally  $q_0(h) < q_1 < q_2(h)$ , so the finite cutoffs inherit the ordering, and if  $q_j(h) \geq \lambda - 1$  then the same holds for every larger breakpoint, so all higher regions are absent as well.  $\square$

*Proof of Lemma 5.* The fixed- $\tau$  problem has a single aggregate inequality constraint, pre-review obedience, together with the pointwise feasibility restrictions on the review menu. Since  $\tau < T$  and  $h < T$ , Assumption 4 and Lemma 2(ii) give  $M\bar{\Delta}(t) > ct$  for  $t \in \{\tau, h\}$ , so full effort is strictly feasible in each phase and the usual constraint qualification holds. The Kuhn–Tucker conditions therefore yield a supporting multiplier  $\lambda^* \geq 0$  such that an optimal menu  $a^*$  maximizes  $\mathcal{L}(a; \lambda^*, \tau)$  over the pointwise feasible actions, satisfies pre-review obedience, and obeys complementary slackness.

Because the Lagrangian is additively separable across review realizations, its maximization at the supporting multiplier can be performed pointwise in  $x$ . Lemma 4 reduces the pointwise choice to the four endpoint actions, and Proposition 2 then implies that an optimal menu can be chosen to have the stated ordered form.

It remains to establish when the multiplier is strictly positive. Suppose  $\lambda^* = 0$ . Then  $Q_0(x) = -1$  for every  $x$  by (15), and the four values in (11) reduce to

$$v_B = 0, \quad v_D = (1 - c)h - r_D(h), \quad v_E = (1 - c)h - r_E(h), \quad v_G = -M.$$

Since  $r_E(h) > r_D(h)$ , only  $A_B$  and  $A_D$  can maximize the Lagrangian, which is Corollary 2. If

$(1 - c)h > r_D(h)$  then  $A_D$  is the unique maximizer at every  $x$ , so the agent receives the constant rent  $r_D(h)$  and

$$\int_{\mathbb{R}} r_D(h) [g_1^\tau(x) - g_0^\tau(x)] dx = 0.$$

If instead  $(1 - c)h < r_D(h)$  then  $A_B$  is the unique maximizer at every  $x$ , the rent is identically zero, and the same conclusion holds. Either case contradicts pre-review obedience, since  $c\tau > 0$ . Hence, outside the knife edge  $(1 - c)h = r_D(h)$ , any supporting multiplier satisfies  $\lambda^* > 0$ , and complementary slackness implies that pre-review obedience binds.

Finally, when  $(1 - c)h = r_D(h)$  the actions  $A_B$  and  $A_D$  are tied at  $\lambda^* = 0$  for every review realization. The principal can then assign these two actions differently across review states, varying the agent's continuation rent and supplying pre-review incentives without sacrificing post-review surplus. A zero multiplier may therefore support the optimum in this knife-edge case, and the strict-positivity conclusion need not hold.  $\square$

*Proof of Lemma A.4.* Let  $\lambda'' > \lambda'$ , let  $(r'(x), s'(x))$  and  $(r''(x), s''(x))$  be pointwise Lagrangian maximizers for  $\lambda'$  and  $\lambda''$ , and write

$$d_\tau(x) = 1 - \frac{g_0^\tau(x)}{g_1^\tau(x)}.$$

Pointwise optimality gives

$$\begin{aligned} s''(x) + \lambda'' d_\tau(x) r''(x) &\geq s'(x) + \lambda'' d_\tau(x) r'(x), \\ s'(x) + \lambda' d_\tau(x) r'(x) &\geq s''(x) + \lambda' d_\tau(x) r''(x). \end{aligned}$$

Adding the two inequalities yields  $(\lambda'' - \lambda') d_\tau(x) [r''(x) - r'(x)] \geq 0$ , hence  $d_\tau(x) [r''(x) - r'(x)] \geq 0$ . Multiplying by  $g_1^\tau(x)$  and integrating,

$$I(\lambda''; \tau) - I(\lambda'; \tau) = \int_{\mathbb{R}} g_1^\tau(x) d_\tau(x) [r''(x) - r'(x)] dx \geq 0.$$

The inequality is strict under the condition stated in the lemma.  $\square$

*Proof of Proposition A.1.* By (A.8),  $k_1(\lambda)$  is finite exactly when  $\lambda > \lambda_2$  and  $k_2(\lambda)$  exactly when  $\lambda > \lambda_3$ . Hence at  $\lambda = \lambda_2$  both upper cutoffs are infinite and at  $\lambda = \lambda_3$  the good-stop cutoff is infinite; substituting into (A.10) and deleting the vanishing terms gives the displayed expressions for  $\mathcal{I}_2$  and  $\mathcal{I}_3$ .

By Lemma 5 the supporting multiplier solves  $I(\lambda; \tau) = c\tau$ , and by Lemma A.4 the supply  $I(\cdot; \tau)$  is weakly increasing. Therefore  $\lambda \leq \lambda_2$  if and only if  $c\tau \leq \mathcal{I}_2(\tau)$ , and  $\lambda \leq \lambda_3$  if and only if  $c\tau \leq \mathcal{I}_3(\tau)$ , which together with Proposition 2 gives (A.13). Since  $\lambda_2 < \lambda_3$ , monotonicity also gives  $\mathcal{I}_2(\tau) \leq \mathcal{I}_3(\tau)$ , strictly at a regular solution. At either equality the new cutoff is infinite, so the newly appearing region has zero probability.  $\square$

## C Proofs for Section 5

### C.1 Proof of Proposition 3

*Proof of Proposition 3.* Apply the constrained envelope theorem to the optimized Lagrangian in (22). Holding the supporting multiplier fixed, differentiation of the terms outside the expectation gives  $1 - \lambda^*c$ .

For the statewise value  $\mathcal{V}(q, h)$ , a change in  $\tau$  operates through two channels. First, it changes the review index  $\tilde{Q}_\lambda(Z; \tau)$ . The resulting contribution is the change in that index multiplied by its marginal value,  $\partial\mathcal{V}/\partial q$ . Second, because  $h = T - \tau$ , delaying the review shortens the continuation horizon one-for-one, producing the term  $-\partial\mathcal{V}/\partial h$ . Thus, holding  $\lambda$  fixed, the chain rule gives the expectation in (23).

Finally, the statewise envelope theorem implies (24): since the slope of each statewise action value with respect to  $q$  is the rent delivered by that action, the derivative of the maximized value with respect to  $q$  is the rent delivered by the selected action. Evaluating all terms at  $\lambda = \lambda^*$  yields (23).  $\square$

### C.2 Proof of Lemma 6

*Proof of Lemma 6.* For interior review dates, continuity follows from the implicit-function characterization of  $z_L(h)$  and  $z_H(h)$ , continuity of the review-action set, and the maximum theorem applied to the fixed-review linear program. Changes in the number of review regions may create changes in the formula used to compute the value, but not jumps in the value itself.

Consider first  $\tau \downarrow 0$ . The largest principal payoff available from a second-phase action is  $s_L(h) = (1 - c)h - r_L(h)$  for all sufficiently small  $\tau$ , because profitability of the no-review contract implies  $s_L(T) = V^{\text{NR}}(T; c, M) > 0$ . Hence

$$V^R(\tau) \leq \tau + s_L(T - \tau),$$

whose limit is  $s_L(T) = V^{\text{NR}}(T; c, M)$ .

For a matching lower bound, assign the low-rent continuation action  $A_L$  at every review state except on a sufficiently high upper tail, on which the principal assigns the good stop  $A_G$ . The utility increment on that tail is  $M - r_L(h)$ . Choose the hard upper-tail cutoff so that

$$[M - r_L(h)]D(\tau, z) = c\tau.$$

For sufficiently small  $\tau$ , such a cutoff exists, and the work probability of the reward region converges to zero. The resulting payoff converges to  $s_L(T)$ . Therefore

$$\lim_{\tau \downarrow 0} V^R(\tau) = V^{\text{NR}}(T; c, M).$$

Now let  $\tau \uparrow T$ , so  $h \downarrow 0$ . A menu using only  $A_B$  and  $A_G$  is exactly a bounded upper-tail bonus

based on  $X_\tau$ . Choosing its hard threshold to satisfy

$$M D(\tau, z) = c\tau$$

gives payoff  $V^{\text{NR}}(\tau; c, M)$ , which converges to  $V^{\text{NR}}(T; c, M)$ .

For the reverse inequality, the full-effort continuation frontier converges to the no-effort frontier. In particular,

$$(1 - c)h \longrightarrow 0, \quad r_L(h) \longrightarrow 0, \quad r_H(h) \longrightarrow M.$$

Thus the complete review-action set converges to

$$\{(r, -r) : r \in [0, M]\}.$$

The review signal converges to the terminal signal  $N(T, T)$ , and the fixed-review problem converges to the no-review bounded-payment problem. This proves

$$\lim_{\tau \uparrow T} V^R(\tau) = V^{\text{NR}}(T; c, M).$$

The remaining conclusions follow from continuity on the compactified interval  $[0, T]$ .  $\square$

### C.3 Proof of Proposition 4

*Proof of Proposition 4.* We prove the three claims in turn.

For the first claim, write  $h = T - \tau$  and use the low-rent continuation action  $A_L(h)$  as a baseline. For all sufficiently small  $\tau$ , this action uniquely maximizes the principal's post-review payoff. Compactness of the two continuation frontiers and strict profitability at  $h = T$  imply that there is a constant  $\kappa > 0$  such that every feasible review action  $(r, s)$  satisfies

$$s_L(h) - s \geq \kappa |r - r_L(h)|. \tag{A.15}$$

Let

$$L_\tau(x) = 1 - \frac{g_0^\tau(x)}{g_1^\tau(x)}.$$

Because a constant review utility provides no incentives, pre-review obedience can be written as

$$\mathbb{E}_1 \left[ (r(X_\tau) - r_L(h)) L_\tau(X_\tau) \right] \geq c\tau. \tag{A.16}$$

Under work,  $X_\tau = \tau + \sqrt{\tau}Z$  and

$$L_\tau(X_\tau) = 1 - \exp \left( -\frac{\tau}{2} - \sqrt{\tau}Z \right).$$

A standard Gaussian tail decomposition gives

$$\mathbb{E}_1 [|r(X_\tau) - r_L(h)|] \geq K \frac{\sqrt{\tau}}{\sqrt{\log(1/\tau)}} \quad (\text{A.17})$$

for some  $K > 0$  and all sufficiently small  $\tau$ . To see the order, split at

$$|Z| = \sqrt{8 \log(1/\tau)}.$$

On the central event,  $|L_\tau|$  is of order  $\sqrt{\tau \log(1/\tau)}$ ; under both work and shirking, the complementary Gaussian tails contribute  $o(\tau)$ . Since review utilities are bounded by  $M$ , (A.16) then implies (A.17).

Combining (A.15) and (A.17) yields

$$V^R(\tau) \leq \tau + s_L(T - \tau) - \kappa K \frac{\sqrt{\tau}}{\sqrt{\log(1/\tau)}}.$$

Since

$$\tau + s_L(T - \tau) = V^{\text{NR}}(T; c, M) + O(\tau),$$

(28) follows.

For the second claim, set  $h = T - \tau$ . As  $h \downarrow 0$ , the work-inducing continuation frontier differs from the no-effort frontier by a vertical surplus of only  $(1 - c)h$ . Moreover,

$$r_L(h) = O\left(\frac{\sqrt{h}}{\sqrt{\log(1/h)}}\right), \quad M - r_H(h) = O\left(\frac{\sqrt{h}}{\sqrt{\log(1/h)}}\right).$$

The continuation actions can therefore improve on the no-effort support function only on likelihood-index intervals whose widths are  $O((\sqrt{h \log(1/h)}))$ . Since the density of the review signal is bounded, their total incremental value is  $o(h)$ . Hence

$$V^R(T - h) = V^{\text{NR}}(T - h; c, M) + o(h).$$

This proves (29).

It remains to differentiate the no-review value. Implicit differentiation of

$$M \left[ \Phi(z_L(T)) - \Phi(z_L(T) - \sqrt{T}) \right] = cT$$

gives

$$\frac{dz_L(T)}{dT} = \frac{c/M - \phi(z_L(T) - \sqrt{T})/(2\sqrt{T})}{\phi(z_L(T)) - \phi(z_L(T) - \sqrt{T})}.$$

Differentiating  $V^{\text{NR}}(T; c, M) = T - M\Phi(z_L(T))$  and using (27) gives (30) and (31). If this derivative is negative, then  $V^R(T - h) > V^R(T)$  for every sufficiently small  $h > 0$ , proving (32).

For the tight-cap limit, let  $a = \sqrt{T}/2$ . As  $M \downarrow \underline{M}(T, c)$ , the two no-review roots merge and

$$z_L(T) \uparrow a, \quad \ell_T \uparrow 1.$$

The numerator in (31) converges to

$$c \left[ 1 - \frac{a\phi(a)}{2\Phi(a) - 1} \right],$$

which is strictly positive because

$$2\Phi(a) - 1 - a\phi(a) > 0 \quad \text{for every } a > 0. \quad (\text{A.18})$$

Indeed, the expression on the left of (A.18) is zero at  $a = 0$  and has derivative  $(1 + a^2)\phi(a) > 0$ . The denominator in (31) converges to zero from above, proving (33).

Finally, as  $M \rightarrow \infty$ , the hard root satisfies  $z_L(T) \rightarrow -\infty$  by the proof of Lemma 3; hence  $\ell_T \rightarrow 0$  and  $M\phi(z_L(T) - \sqrt{T}) \rightarrow 0$ . Equation (31) therefore yields (34).  $\square$

#### C.4 Proof of Corollary 3

*Proof of Corollary 3.* At  $M = \underline{M}(T, c)$ , the no-review value is

$$T \left[ 1 - \frac{c\Phi(\sqrt{T}/2)}{2\Phi(\sqrt{T}/2) - 1} \right],$$

which is nonnegative exactly under (35). For caps just above the feasibility boundary, the hard-threshold implementation cost falls and the no-review payoff is nonnegative, strictly positive except possibly at the boundary itself. Proposition 4 implies that  $V_-^{R'}(T) < 0$  throughout a sufficiently small right neighborhood of  $\underline{M}(T, c)$ . Moving the review slightly to the left of  $T$  then strictly raises the principal's payoff.  $\square$

#### C.5 Proof of Proposition 5

*Proof of Proposition 5.* At the boundary,

$$\frac{\overline{D}(T)}{T} = \frac{c}{M_0}.$$

Since  $\overline{D}(t)/t$  is strictly decreasing by Lemma 2(ii), every  $t < T$  satisfies

$$\frac{\overline{D}(t)}{t} > \frac{c}{M_0},$$

which gives (36) for  $t \in \{\tau, h\}$ .

To obtain the second claim, choose  $\tau_0 > 0$  small and write  $h_0 = T - \tau_0$ . At  $M = M_0$  the



continuation problem at  $h_0 < T$  is strictly feasible, so its minimum work-inducing rent  $r_L(h_0)$  is well defined and strictly below  $M_0$ . At the boundary  $h = T$ , the two threshold roots coincide and the corresponding rent is  $M_0\Phi(-\sqrt{T}/2)$ ; hence the available rent spread converges to  $M_0\Phi(\sqrt{T}/2) > 0$  as  $h_0 \uparrow T$ . Since  $\bar{D}(\tau)/\tau \rightarrow \infty$  as  $\tau \downarrow 0$ ,  $\tau_0$  can therefore be chosen small enough that

$$[M_0 - r_L(h_0)]\bar{D}(\tau_0) > c\tau_0. \quad (\text{A.19})$$

Both continuation feasibility at  $h_0$  and (A.19) are strict, so by continuity they continue to hold for all caps in a sufficiently small interval  $(M_0 - \varepsilon, M_0]$ .

For such a cap, use the minimum-rent full-effort continuation contract after review outcomes  $X_{\tau_0} < \tau_0/2$ , and use the good-stop action with rent  $M$  after  $X_{\tau_0} \geq \tau_0/2$ . The spread in continuation utility between the two regions is  $M - r_L(h_0)$ , and the midpoint cutoff maximizes the difference between the work and shirk probabilities. The date-0 incentive spread is therefore

$$[M - r_L(h_0)]\bar{D}(\tau_0) > c\tau_0,$$

so first-phase work is induced. In the lower region the continuation contract induces full effort, while in the upper region the project is stopped. The lower region has strictly positive probability under work. Finally,  $M < M_0 = \underline{M}(T, c)$  makes full effort infeasible without review by Lemma 2(i).  $\square$

## C.6 Proof of Proposition 6

*Proof of Proposition 6.* At a two-region point, let  $z_0$  denote the standardized cutoff under first-phase work. Using

$$s_L(h) = (1 - c)h - r_L(h),$$

the optimized value and binding first-phase obedience condition are

$$\hat{V}_2(\tau, h) = \tau + s_L(h)\Phi(z_0), \quad r_L(h)D(\tau, z_0) = c\tau. \quad (\text{A.20})$$

Let  $\lambda$  be the supporting multiplier on the obedience constraint. Because the high-rent action  $A_H$  is absent, the regime characterization in Proposition 2 and (A.11) gives  $\lambda \leq 1$ . At the regular points covered by the lemma, Lemma 5 also gives  $\lambda > 0$ .

The constrained envelope theorem applied to (A.20) yields

$$\hat{V}_{2,\tau} = 1 + \lambda[r_L(h)D_\tau(\tau, z_0) - c]$$

and

$$\hat{V}_{2,h} = s'_L(h)\Phi(z_0) + \lambda r'_L(h)D(\tau, z_0).$$

Since  $s'_L(h) = 1 - c - r'_L(h)$ , subtraction gives

$$\begin{aligned} G_2(\tau, h) &= 1 - \Phi(z_0) + c[\Phi(z_0) - \lambda] \\ &\quad + r'_L(h)[\Phi(z_0) - \lambda D(\tau, z_0)] + \lambda r_L(h) D_\tau(\tau, z_0). \end{aligned} \quad (\text{A.21})$$

We establish two bounds and then use them in this expression.

First, the minimum continuation rent cannot fall too quickly with the continuation horizon. Recall that  $z_L(h) < \sqrt{h}/2$  solves

$$MD(h, z_L(h)) = ch$$

and that

$$r_L(h) = M\Phi(z_L(h) - \sqrt{h}).$$

Implicit differentiation of the binding incentive condition gives

$$z'_L(h) = \frac{c/M - \phi(z_L(h) - \sqrt{h})/(2\sqrt{h})}{\phi(z_L(h)) - \phi(z_L(h) - \sqrt{h})}.$$

Therefore

$$r'_L(h) = \frac{\phi(z_L(h) - \sqrt{h})}{\phi(z_L(h)) - \phi(z_L(h) - \sqrt{h})} \left[ c - \frac{M\phi(z_L(h))}{2\sqrt{h}} \right]. \quad (\text{A.22})$$

The inequality  $r'_L(h) > -c/2$  is equivalent, after substituting  $c = MD(h, z_L(h))/h$ , to

$$D(h, z_L(h)) > \sqrt{h} \frac{\phi(z_L(h))\phi(z_L(h) - \sqrt{h})}{\phi(z_L(h)) + \phi(z_L(h) - \sqrt{h})}. \quad (\text{A.23})$$

To verify this inequality, note that

$$D(h, z_L(h)) = \int_{z_L(h) - \sqrt{h}}^{z_L(h)} \phi(t) dt.$$

Because  $z_L(h) < \sqrt{h}/2$ ,  $\phi(z_L(h)) > \phi(z_L(h) - \sqrt{h})$ . Unimodality of the normal density then implies that its minimum on the interval of integration is  $\phi(z_L(h) - \sqrt{h})$ . Hence

$$D(h, z_L(h)) > \sqrt{h} \phi(z_L(h) - \sqrt{h}),$$

which is strictly stronger than (A.23). Thus

$$r'_L(h) > -\frac{c}{2}. \quad (\text{A.24})$$

Second, the direct effect of a longer first phase on the Gaussian incentive spread is sufficiently large. Holding  $z_0$  fixed,

$$D_\tau(\tau, z_0) = \frac{\phi(z_0 - \sqrt{\tau})}{2\sqrt{\tau}}.$$

Using the binding obedience condition in (A.20),

$$r_L(h)D_\tau(\tau, z_0) = \frac{c\sqrt{\tau}\phi(z_0 - \sqrt{\tau})}{2D(\tau, z_0)}. \quad (\text{A.25})$$

We claim that

$$D(\tau, z_0) < \sqrt{\tau} \frac{\phi(z_0 - \sqrt{\tau})}{\Phi(z_0 - \sqrt{\tau})}. \quad (\text{A.26})$$

If  $z_0 - \sqrt{\tau} \geq 0$ , the normal density is decreasing throughout the interval  $[z_0 - \sqrt{\tau}, z_0]$ , so its average over that interval is strictly below  $\phi(z_0 - \sqrt{\tau})$ , which is itself strictly below the right-hand ratio in (A.26). If  $z_0 - \sqrt{\tau} < 0$ , the average density is at most  $\phi(0)$ , whereas the normal reverse hazard satisfies

$$\frac{\phi(z_0 - \sqrt{\tau})}{\Phi(z_0 - \sqrt{\tau})} \geq \frac{\phi(0)}{\Phi(0)} = 2\phi(0).$$

Thus (A.26) holds in either case. Combining it with (A.25) gives

$$r_L(h)D_\tau(\tau, z_0) > \frac{c}{2}\Phi(z_0 - \sqrt{\tau}). \quad (\text{A.27})$$

We now return to (A.21). Since  $0 < \lambda \leq 1$ ,

$$\Phi(z_0) - \lambda D(\tau, z_0) \geq \Phi(z_0) - D(\tau, z_0) = \Phi(z_0 - \sqrt{\tau}) > 0.$$

Using (A.24), we therefore obtain

$$\begin{aligned} G_2(\tau, h) &> 1 - \Phi(z_0) + \frac{c}{2}\Phi(z_0) \\ &\quad + \lambda \left[ r_L(h)D_\tau(\tau, z_0) - c \left( 1 - \frac{D(\tau, z_0)}{2} \right) \right]. \end{aligned} \quad (\text{A.28})$$

If the expression in square brackets is nonnegative, the right-hand side is strictly positive. If it is negative, then  $\lambda \leq 1$  implies that the right-hand side of (A.28) is strictly greater than

$$(1 - c)[1 - \Phi(z_0)] + r_L(h)D_\tau(\tau, z_0) - \frac{c}{2}\Phi(z_0 - \sqrt{\tau}).$$

The first term is nonnegative because  $c \in (0, 1)$ , and the remaining two terms have a strictly positive sum by (A.27). Hence  $G_2(\tau, h) > 0$  in either case. Imposing  $h = T - \tau$  gives  $dV^R/d\tau = G_2 > 0$  throughout every smooth two-region segment, which rules out a smooth interior optimum in that regime.  $\square$

## C.7 Proof of Proposition 7

*Proof of Proposition 7.* By (30),

$$V_-^{Rl}(T) = 1 - \lambda_T^{\text{NR}}c + \lambda_T^{\text{NR}} \frac{M\phi(z_L(T) - \sqrt{T})}{2\sqrt{T}}.$$

The last term is strictly positive. Hence  $V_-^{R'}(T) < 0$  implies

$$\lambda_T^{\text{NR}} > \frac{1}{c}. \quad (\text{A.29})$$

Next consider the boundary between the three- and four-region menus at a continuation horizon  $h > 0$ . By (12),

$$q_2(h) = \frac{(1-c)h}{M - r_H(h)}.$$

Proposition 1 gives  $r_H(h) < M - ch$ , so  $M - r_H(h) > ch$ . Therefore

$$1 + q_2(h) < 1 + \frac{1-c}{c} = \frac{1}{c}. \quad (\text{A.30})$$

By the active-region characterization in Appendix B, the good-stop region is nonempty whenever the supporting multiplier exceeds  $1 + q_2(h)$ .

It remains to compare the fixed-review multiplier near  $T$  with the no-review multiplier. We do this directly from the incentive-supply characterization. For  $\lambda > 1$ , let

$$I^{\text{NR}}(\lambda; T) = MD(T, z_1(\lambda, T)),$$

where  $z_1(\lambda, T)$  is obtained from (A.9) using  $q_1 = 0$ . This is the incentive spread supplied by the no-review upper-tail contract supported by  $\lambda$ . By (27),

$$z_1(\lambda_T^{\text{NR}}, T) = z_L(T),$$

and therefore

$$I^{\text{NR}}(\lambda_T^{\text{NR}}; T) = cT. \quad (\text{A.31})$$

The function  $I^{\text{NR}}(\lambda; T)$  is strictly increasing at  $\lambda_T^{\text{NR}}$ .

Now fix any two multipliers

$$\frac{1}{c} < \lambda_- < \lambda_T^{\text{NR}} < \lambda_+.$$

For  $\tau$  sufficiently close to  $T$ , (A.30) implies that the four-region formula (A.10) applies at both  $\lambda_-$  and  $\lambda_+$ . As  $h = T - \tau \downarrow 0$ , the endpoint limits in the proof of Lemma 6 give

$$r_L(h) \longrightarrow 0, \quad r_H(h) \longrightarrow M.$$

Hence the first and third terms of (A.10) vanish, while the middle term converges to the no-review incentive spread:

$$I(\lambda; \tau) \longrightarrow I^{\text{NR}}(\lambda; T) \quad \text{for } \lambda \in \{\lambda_-, \lambda_+\}. \quad (\text{A.32})$$

Strict monotonicity and (A.31) imply

$$I^{\text{NR}}(\lambda_-; T) < cT < I^{\text{NR}}(\lambda_+; T).$$

Therefore, for every  $\tau$  sufficiently close to  $T$ ,

$$I(\lambda_-; \tau) < c\tau < I(\lambda_+; \tau).$$

Lemma A.4 and (18) then imply that every supporting multiplier satisfies

$$\lambda_- < \lambda^*(\tau) < \lambda_+.$$

Since  $\lambda_- > 1/c$ , there exists  $\delta_1 > 0$  such that, for every  $\tau \in (T - \delta_1, T)$ ,

$$\lambda^*(\tau) > \frac{1}{c} > 1 + q_2(T - \tau).$$

Proposition 2 therefore implies that all four review regions are nonempty throughout this terminal neighborhood.

Finally, Lemma 6 gives  $V^R(T) = V^{\text{NR}}(T; c, M)$ . Since  $V_-^{R'}(T) < 0$ , there exists  $\delta_2 > 0$  such that

$$V^R(\tau) > V^R(T) = V^{\text{NR}}(T; c, M)$$

for every  $\tau \in (T - \delta_2, T)$ . Taking  $\delta = \min\{\delta_1, \delta_2\}$  proves the first two claims. The final statement follows from (33), which makes the terminal derivative negative for every sufficiently tight feasible cap under the maintained profitability condition.  $\square$